VDD

VSS

INPUT FROM SYNTHESIS
- Gate Level Netlist
- Libraries
- Constraints (SDC)

DESIGN UNDERSTANDING
- Block Analysis
- I/O Analysis
- Power Intent

POWER PLANNING
- Power Network
- IR Drop Control
- EM Reliability

FLOORPLAN CREATION
- Die & Core Area
- I/O Placement
- Power Plan

PLACEMENT
- Macro Placement
- Blockage / Keepout
- Channel Planning

CHECKS & VALIDATION
- DRC / LVS
- IR Drop
- EM / Antenna

READY FOR NEXT STAGE

VDD

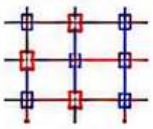
VSS

PD POWER PLANNING

POWER PLANNING

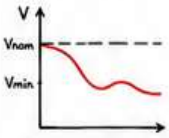
Power Planning builds a robust Power Delivery Network (PDN) that ensures stable voltage, low IR Drop, high reliability and optimum performance for the entire chip.

POWER NETWORK CONCEPT



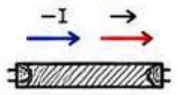
- Power Rings
- Power Straps
- Power Mesh
- Decap Cells

IR DROP CONTROL



- Minimize IR Drop
- Uniform Voltage
- Better Performance

ELECTROMIGRATION (EM)



- High Current Density causes EM
- Ensure EM Reliability
- Follow Foundry Rules

POWER INTEGRITY

- Stable Voltage
- Low Noise
- Reliable Operation

SIGNOFF READINESS

- IR Drop Check
- EM Check
- DRC / LVS
- Antenna Check

POWER STRUCTURES

- Power Ring (VDD)
- Power Ring (VSS)
- Power Strap (V)
- Power Strap (H)
- Decap Cell

DECAP CELLS

- Reduce Local Noise
- Supply Transient Support
- Improve Stability

POWER EFFICIENCY

- Efficient PDN
- Optimized Area
- Better Yield

RELIABILITY

- IR Drop Safe
- EM Safe
- Robust PDN

POWER PLANNING FLOW

1. POWER INTENT & REQUIREMENTS	2. POWER NETWORK DESIGN	3. DECAP PLACEMENT STRATEGY	4. ANALYSIS & OPTIMIZATION	5. VALIDATION & SIGNOFF
• Voltage Levels • Current Demand • Power Domains	• Rings & Straps • Mesh Planning • Via Strategy	• Cell Selection • Placement Rules • Density Planning	• IR Drop Analysis • EM Analysis • Optimization	• DRC / LVS • IR / EM / ANT • Final Signoff

POWER PLANNING GOALS

Stable Voltage | Low IR Drop | EM Reliability | Noise Reduction | High Performance | Better Yield | Power Efficiency

KEY ENGINES IN POWER PLANNING

IR DROP ENGINE	EM ENGINE	POWER GRID ENGINE	DECAP ENGINE	DFM ENGINE	REPORTING ENGINE
• Voltage Map • Drop Analysis • Hotspot Fixing	• Current Density • EM Check • Via Insertion	• Grid Generation • Strap Sizing • Connectivity	• Decap Insertion • Optimization • Noise Reduction	• DRC Clean • Manufacturable • Rule Compliant	• Coverage • Violations • Summary

A Strong Power Plan Today
Leads to a Stable Chip Tomorrow.
Plan Right. Power Right. Perform Right.

Author :
Konda Tharun Vivekananda

VDD

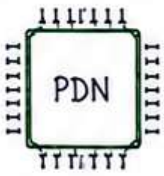
POWER RING

POWER RING

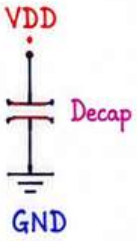
VSS

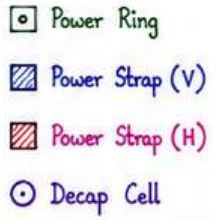
VDD

VSS



PD Power Planning

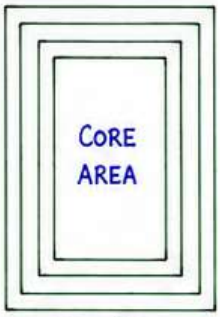




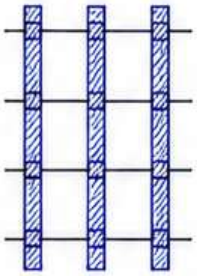
• • • INDEX • • •

S.No.	Topic	Page No.
1.	Introduction to PD Power Planning	
2.	PDN Structure – Details & Hierarchy	
3.	Power Planning Flow – Detailed	
4.	Power Delivery Challenges (Nodes / Bumps)	
5.	Power Network Concept	
6.	Power Planning Components	
7.	Power Network Hierarchy	
8.	IR Drop Concept	
9.	IR Drop Effects on Timing	
10.	Electromigration (EM)	
11.	Power Planning Best Practices	
12.	Common Mistakes in Power Planning	
13.	Viva / Interview Points – Part 1	
14.	Important Metrics & Parameters	
15.	PD Power Planning Summary Points	
16.	Decoupling Capacitors (DEcap Cells) – Definition, Function, Benefits	
17.	All Guidelines to Follow in All Stages of Power Planning	

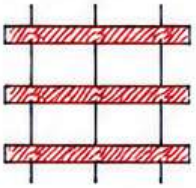
Power Rings



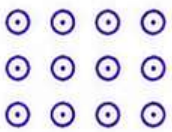
Power Straps (V)



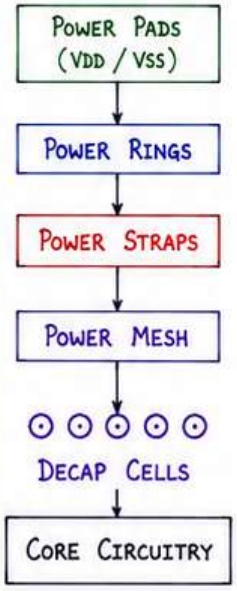
Power Straps (H)



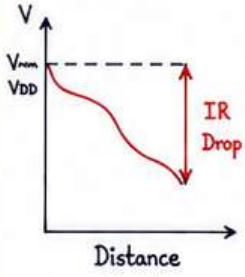
Decap Cells



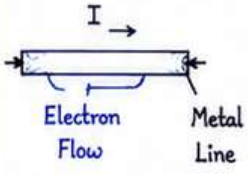
PDN Hierarchy



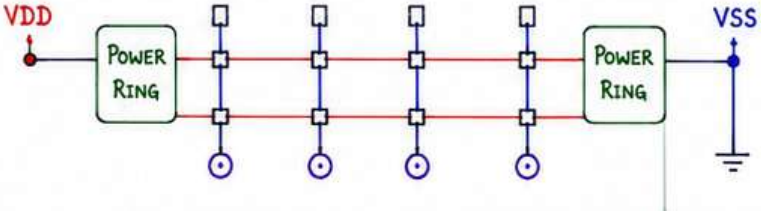
IR Drop

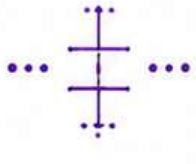


EM Concept



Author :
Konda Tharun Vivekananda





PD Power Planning

1. Introduction

Power Planning is a key step in Physical Design that builds a robust Power Delivery Network (PDN) to supply power (V_{DD}) and ground (V_{SS}) to all parts of the chip with minimal IR Drop and EM issues. It works closely with Floorplan and impacts timing, reliability and manufacturability.

2. Why Power Planning is Critical ?

- Modern chips have high current demand.
- Poor PDN → IR Drop → Performance loss.
- Voltage droop → Timing failures / Resets.
- Electromigration (EM) risk increases.
- Ensures chip reliability and lifetime.
- Supports multi-voltage / multi-domain designs.
- Reduces simultaneous switching noise (SSN).
- Helps meet sign-off for IR Drop and EM.

3. Objectives of Power Planning

1. Deliver clean, stable power with minimal IR Drop.
2. Handle peak current demand under all modes (Worst Case).
3. Avoid EM failures in power network.
4. Ensure reliable operation across PVT (Process, Voltage, Temperature).
5. Support multiple voltage domains effectively.
6. Provide good scalability and manufacturability.
7. Minimize chip area overhead due to PDN.
8. Ensure good noise immunity (SSN / Ldi/dt).

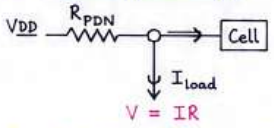
Key Point

A good PDN = Low IR Drop + No EM + Reliable across PVT + Efficient area usage.
Power Planning is the backbone of chip reliability.

Practical Observation

- Wider straps ↓ IR Drop but ↑ area.
- More decaps near high switching blocks improves local stability.
- Poor power plan → Timing violations, EM failures, chip may not boot.
- Early power planning reduces rework and ECOs.
- Always verify IR Drop and EM after PDN build.

8. IR Drop Concept



$IR\ Drop = I_{load} \times R_{PDN}$

High I_{load} or High R_{PDN}
→ High IR Drop
→ Voltage droop

9. IR Drop Effect on Timing

Effect	Impact
Lower VDD	Slower cells
Delay increase	Setup violation risk
Noise increase	Reliability issues / Resets

Target

- IR Drop < 5% of V_{DD} (typ.)
- At worst-case current
- Across PVT corners

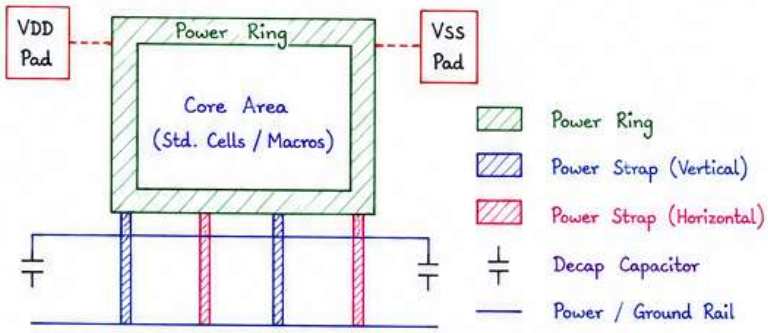
4. Power Delivery Challenge

Power Source (Pads / Bumps)
↓
Pkg. / Board
↓
On-Chip PDN
↓
Standard Cells / IPs / Macros

Challenges :

- High current density
- IR Drop
- EM
- Voltage noise
- Multiple voltage domains
- Area overhead
- Simultaneous switching noise (SSN)

5. Power Network Concept



Power flows : Pads → Rings → Straps → Mesh → Cells
Decaps reduce local IR Drop and supply noise.

6. Power Planning Components

Component	Function	Typical Location	Notes
Pads / Bumps	Entry point of power	Chip periphery	High current capability
Power Rings	Surround core, distribute	Periphery of core	Usually on M5/M6
Power Straps	Carry current deep inside	Vertical / Horizontal	Wide metals (M3–M6)
Power Mesh	Uniform distribution	Inside core	Grid of straps
Decap Capacitors	Supply local charge	Near high activity	Reduces IR drop

7. Power Network Hierarchy

Pads / Bumps → Power Rings → Power Straps → Power Mesh → Cells / Loads

- Higher levels (Rings, Straps) handle bulk current.
- Lower levels (Mesh, Decaps) handle local distribution.

10. Electromigration (EM)

High current density causes metal atom movement → Voids → Open circuits. Check using EM rules.

EM Rule of Thumb :

Current Density (J) < J_{max}

$J = I / Area$

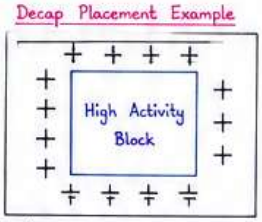
Factors Increasing EM Risk :

- High current
- High temperature
- Small metal width
- Long metal length

11. Power Planning Best Practices

- ✓ Start power planning early (with floorplan).
- ✓ Use wide metals for straps and rings.
- ✓ Keep strap pitch balanced in both directions.
- ✓ Place decaps near high switching blocks.
- ✓ Verify IR Drop and EM after each iteration.
- ✓ Follow foundry PDN guidelines and rules.

Decap Placement Example



⏚ = Decap

12. Common Mistakes

- ✗ Ignoring early power planning.
- ✗ Using narrow straps.
- ✗ Overlooking decap placement.
- ✗ Ignoring multi-voltage domains.
- ✗ Not verifying IR Drop / EM early.

13. Viva / Interview Points

Q1. What is Power Planning ?
→ Designing PDN to deliver clean power (V_{DD}) and ground (V_{SS}) to all parts of the chip.

Q2. Why is IR Drop important?
→ It causes voltage droop and timing failures / resets.

14. Important Metrics

- IR Drop (mV)
- Current Density (mA/μm²)
- EM Check (MTTF)
- Decap Coverage (%)
- Voltage Noise (mV)
- Power Integrity (PI)
- Power Efficiency

15. Summary Points

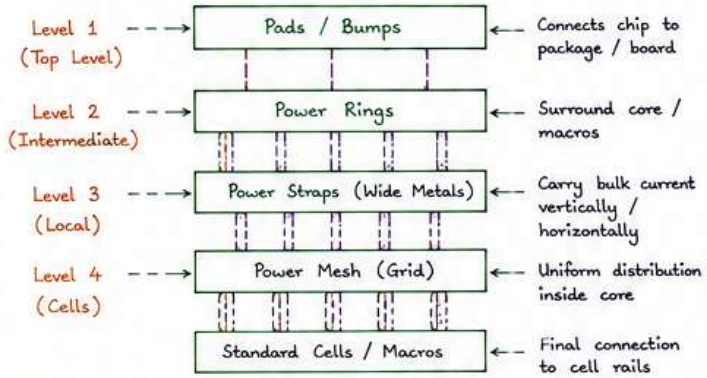
- Power Planning ensures stable and reliable power delivery.
- It reduces IR Drop, EM and voltage noise.
- Good PDN → Better performance and reliability.
- Early and proper power planning is key to successful tape-out.

Author: Konda Tharun Vivekananda

PD Power Planning

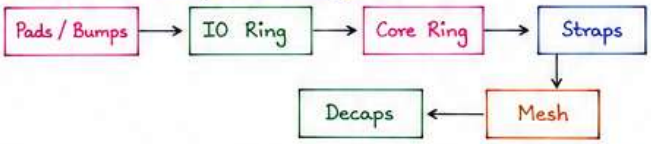
1. PDN Structure – Details

→ The Power Delivery Network (PDN) is a hierarchical network of metals + vias used to distribute power (V_{DD}) and ground (VSS) to all standard cells, macros and IPs.



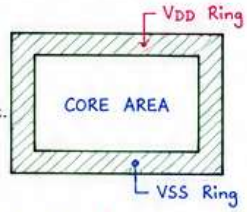
- Many vias are used between levels to reduce resistance and improve current flow capability.

2. Power Planning Hierarchy (Flow)



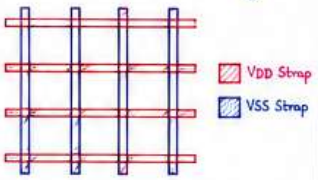
3. Power Ring

- Metal ring around core / macros.
- Provides low resistance path for bulk current.
- Usually on higher metals (M5 – M8).
- Width decided by current requirement.



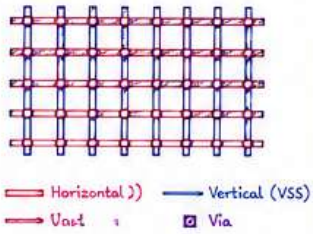
4. Power Straps (Rails)

- Wide metal lines running across core vertically and horizontally.
- Connect power rings to mesh.
- Carry large amount of current.
- Pitch and width depend on IR drop and EM limits.



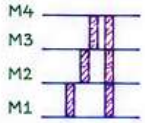
5. Power Mesh (Grid)

- Dense grid of metal lines.
- Connected to straps using vias.
- Distributes power uniformly to all standard cells.
- Usually on lower metals (M1–M4).



6. Via Strategy

- Use multiple vias (via arrays) between layers.
- Via stacking aligned to power straps / mesh.
- Reduces via resistance and improves reliability.

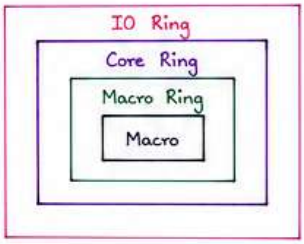


Key Point

A well planned PDN reduces IR Drop, EM and noise, ensures reliability and improves chip performance across all PVT corners.

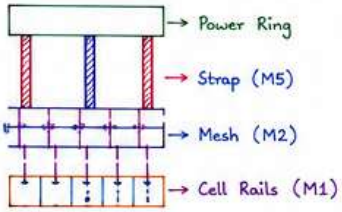
7. Power Ring Types

- Core Ring : Surrounds core area.
- Macro Ring : Surrounds each macro.
- IO Ring : Near IO pads.
- Well Tap Ring (For analog).



8. Power Mesh Connection

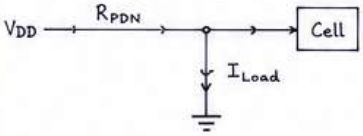
- Straps connect rings to mesh.
- Mesh connects to cell rails.
- Decaps placed between mesh and rails.



9. Power Planning Terminologies

Term	Description
IR Drop	Voltage drop due to current flow in PDN.
EM	Electromigration due to high current density.
SSN	Simultaneous Switching Noise.
Decap	Decoupling Capacitor.
PVT	Process, Voltage, Temperature.
Current Density (J)	I / Area (A / μm^2)
IR Drop (%)	($V_{Nom} - V_{Actual}$) / V_{Nom} × 100

10. IR Drop Concept



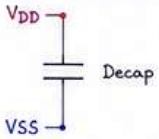
$V_{Drop} = I_{Load} \times R_{PDN}$

IR Drop ↑ if :

- I_{Load} ↑ (High activity)
- R_{PDN} ↑ (Narrow / long path)
- Temperature ↑

11. Decoupling Capacitors (Decaps)

- Provide local charge when current demand is high.
- Reduce IR Drop and supply noise.
- Placed between V_{DD} and VSS.
- Types : MIM, MOM, Metal-Gate.



12. Good PDN vs Poor PDN

Good PDN	Poor PDN
✓ Low IR Drop	× High IR Drop
✓ Uniform power distribution	× Voltage droop
✓ Low EM risk	× Timing failures
✓ Low noise (SSN)	× High EM risk
✓ Stable voltage	× Noise / SSN issues
✓ Better performance	× Performance loss
✓ Reliable operation	× May cause chip failure

13. Practical Guidelines

- Start power planning early (with floorplan).
- Use wide metals for rings and straps.
- Keep straps pitch balanced in both directions.
- Place decaps near high switching / active blocks.
- Verify IR Drop & EM after each iteration.
- Follow foundry PDN guidelines and design rules.

14. Common Issues

- Insufficient strap width
- Too few vias
- Wrong strap pitch
- Missing decaps
- Ignoring IR / EM
- Unbalanced mesh

15. Interview / Viva

Q1. What is PDN ?
→ Network that delivers V_{DD} and VSS to all parts of the chip.

Q2. Why IR Drop occurs ?
→ Resistance in PDN + high current.

Q3. How to reduce IR Drop ?
→ Wider metals, more vias, decaps, balanced straps & mesh.

16. Important Metrics

- IR Drop (mV / %)
- Current Density (mA / μm^2)
- EM Check (MTTF)
- Decap Coverage (%)
- Voltage Noise (mV)
- Power Integrity (PI) Score
- Power Efficiency

17. Summary Points

- PDN is hierarchical network from pads to cell rails.
- Rings, straps and mesh distribute power efficiently.
- Proper power planning reduces IR Drop, EM and noise.
- Decaps improve stability and reliability.
- Good PDN ensures timing closure and robust operation.
- Early and correct power planning is key for successful tape-out.

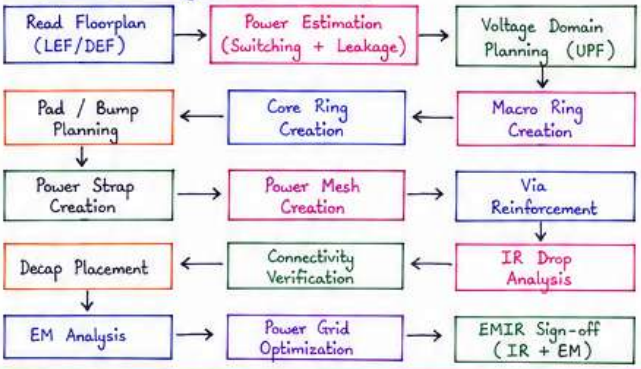
Author: Konda Tharun Vivekananda

PD Power Planning

1. Input Files Needed and Why

Input File	Purpose	Why it is Needed in Power Planning
LEF (Library Exchange Format)	Physical information about cells / macros (ports, pins, blockages, obs, site, layers).	To know pin locations, blockages and keep-outs for ring / strap / mesh creation and connectivity.
DEF (Design Exchange Format)	Floorplan information (die area, core, macros, IOs, placement, rows).	Defines the layout region to build PDN and connect macros, std. cells and IO.
Liberty (.lib)	Cell electrical characteristics (power, leakage, internal pins).	To estimate power, current demand and perform IR Drop & EM analysis.
UPF (Unified Power Format)	Power intent (power nets, supply sets, isolation, domains, isolation, level shifters, retention).	To understand power architecture, voltage domains and connectivity requirements.
Technology Files (Tech LEF, Tech LEF)	Process layers, via defs, min widths, spacing, rules.	To follow manufacturing rules while creating rings, straps, mesh and vias.
Power Intent / CSV / Excel	Power budget, current density, domain info, decap targets.	Guides estimation, strap pitch/width, mesh density & decap requirement.
SDC (Timing Constraints)	Clock, uncertainty, constraints.	Affects activity factor and switching power estimation.
Netlist (Mapped)	Gate level netlist.	Used for power estimation (switching + leakage).
IR Drop / EM Sign-off Rules	Pose values from PDK or company guidelines.	To ensure PDN meets IR Drop and EM limits.

2. Power Planning Flow – Detailed



Goal : Deliver clean, stable power with minimal IR drop and EM issues.
- Meet all sign-off requirements for all PVT corners.

3. Power Estimation Basics

- Total Power = Switching Power (P_{sw}) + Leakage Power (P_{leak})
- $P_{sw} = \Sigma (\alpha \times C \times V^2 \times f)$, $P_{leak} = \Sigma (I_{leak} \times V)$
- Used for deciding strap width / pitch, mesh density, decap count and voltage domain sizing.

Key Point

Accurate power estimation leads to right PDN sizing and prevents late stage IR Drop / EM violations.

Viva / Interview Points

Q1. What are the input files required for Power Planning?
→ LEF, DEF, Liberty, UPF, Tech files, Netlist, SDC, Power budget etc.

Q2. Why is UPF important in Power Planning ?
→ It defines voltage domains, power nets and supply intent to build correct PDN.

Q3. What happens if PDN is not planned properly ?
→ High IR Drop, EM failures, noise issues, timing failures and chip may not boot.

Practical Observation

- More vias → Lower resistance → Better current flow.
- Wider straps → Lower IR Drop but more area.
- Better decap placement near high activity blocks reduces noise and IR Drop.
- Check power grid early before CTS and Routing to avoid rework.

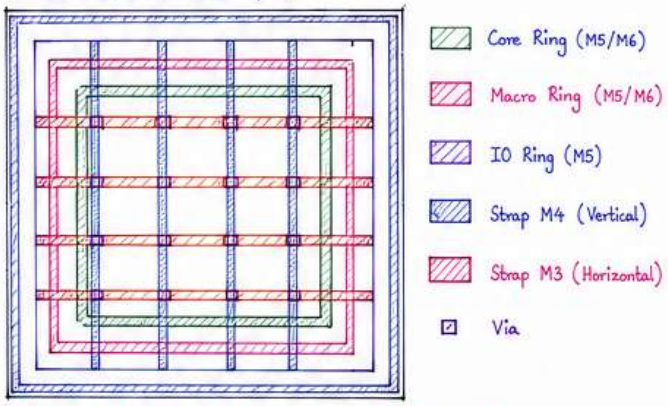
4. Power Grid Layers and Usage

Layer	Direction	Typical Use
M1	Horizontal	Cell rails connection
M2	Vertical	Local mesh / cell connection
M3	Horizontal	Straps (fine pitch)
M4	Vertical	Straps (coarser pitch)
M5 / M6 +	Horizontal / Vertical	Core & Macro rings, global straps, mesh backbone (Low R metal)

5. PDN Structure Summary

Pads / Bumps	Entry point of power from package / board.
Power Rings	High current path around core / macros.
Power Straps	Wide metals running across core to carry bulk current.
Power Mesh	Grid of metals inside core for uniform distribution.
Decaps	Provide local charge, reduce IR Drop and supply noise.
Vias	Connect different metal layers.

6. PDN Top View (Example)



Uniform mesh + proper strap pitch + enough vias → Low IR Drop & EM safe.

7. IR Drop and EM Limits

IR Drop Limit (Typical)	Em Limit (Typical)
• On-chip PDN : 5% – 10% of V_{DD} • At worst-case current • All PVT corners	• Current density < J_{max} • J_{max} = 0.8 ~ 1 MA/cm^2 (Cu) • Based on process & metal layer

8. Power Planning Checklist

- ✓ All voltage domains defined (UPF)
- ✓ Power nets connectivity verified
- ✓ Rings closed and connected
- ✓ Strap width & pitch as per IR/EM
- ✓ Mesh density sufficient
- ✓ Vias adequate and well distributed
- ✓ Decaps placed near high activity blocks
- ✓ IR Drop within limit
- ✓ EM current density within limit
- ✓ Sign-off reports clean (EMIR)

Common Mistakes

- X Late power planning (after placement / CTS).
- X Too narrow straps or less via count.
- X Ignoring macro ring and connections.
- X Wrong strap pitch / imbalance in both directions.
- X No decap placement or poor placement.
- X Not checking IR Drop / EM early.

10. Power Planning Metrics (Important)

- IR Drop (mV / %)
- Current Density (mA/μm^2)
- EM Check (MTTF)
- Decap Coverage (%)
- Voltage Noise (mV)
- Power Integrity (PI) Score
- Power Efficiency (%)

11. Summary / Revision Points

1. Power Planning builds a robust PDN to deliver V_{DD} and Vss to all parts.
2. It includes rings, straps, mesh, vias and decaps.
3. Input files like LEF, DEF, Liberty, UPF and Power budget are required.
4. Aim is to reduce IR Drop, EM and noise while ensuring reliability.
5. Proper grid design ensures timing closure, reduces ECOs and helps successful tape-out.

Author: Konda Tharun Vivekananda

PD Power Planning

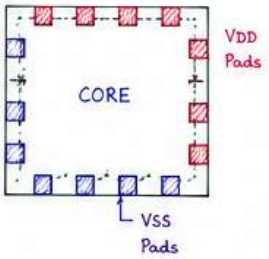
1. Power Pads

(a) Definition

Power Pads are special I/O pads used to bring power (V_{DD}) and ground (VSS) from the package or board to the on-chip Power Delivery Network (PDN). They are placed along the die periphery.

(b) Functions

- Provide low resistance path for high current.
- Distribute V_{DD} and Vss to power rings.
- Support multiple voltage domains.
- Reduce IR Drop and EMC/EM issues.
- Ensure reliable and stable chip operation.



(c) Advantages

- ✓ Easy to connect with package / board.
- ✓ Can use multiple pads for high current.
- ✓ Well understood & widely supported by EDA flows.
- ✓ Flexible placement around die edge.
- ✓ Good for wire-bond based packages.

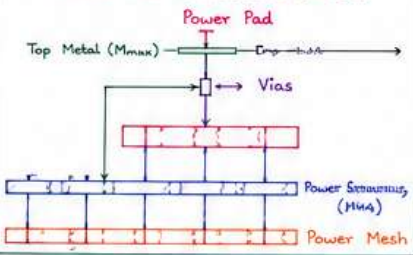
Power Pad Types

Type	Description	Usage
Core VDD Pad	Supplies V_{DD} to core domain	Core power
Core VSS Pad	Supplies Vss to core domain	Core ground
IO V_{DD} Pad	Supplies V_{DD} to IO domain	IO power
IO VSS Pad	Supplies Vss to IO domain	IO ground
Analog Pad	Supplies analog rails (AV_{DD}/AGND)	Analog blocks

(e) Power Pad Placement Guidelines

- Distribute pads uniformly around the die periphery.
- Alternate V_{DD} and VSS pads to reduce loop inductance.
- Place more pads near high current blocks.
- Keep pads connected to power rings using multiple vias.
- Follow foundry recommended pad cell and spacing rules.

Power Pad Connection to PDN



Key Note

More pads
+ proper spacing
+ strong PDN
= Low IR Drop & Better EM Reliability.

3. Pads vs Bumps - Quick Comparison

Parameter	Power Pads	Power Bumps
Connection	Wire bond / Package pins	Flip-chip / Substrate
Current Capacity	Moderate	Very High
Inductance	High (due to bond wires)	Low
Die Area Usage	Consumes periphery	Minimal
Cost	Low	High
Reliability	Good	Better
Typical Use	Low-Medium current chips	High performance SoCs
Technology	Any node	Advanced nodes

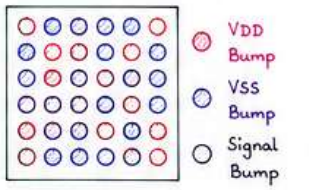
2. Power Bumps

(a) Definition

Power Bumps are flip-chip bumps dedicated to deliver power (V_{DD}) and ground (Vss) between the chip and package substrate.

(b) Functions

- Provide very low inductance & resistance path.
- Handle very high current demand.
- Improve power integrity & reduce IR Drop.
- Reduce package parasitics and improve performance.



(c) Advantages

- ✓ Very low inductance and resistance.
- ✓ Supports very high current density.
- ✓ Better power integrity and performance.
- ✓ Saves die edge area.
- ✓ Ideal for advanced nodes & large SoCs.

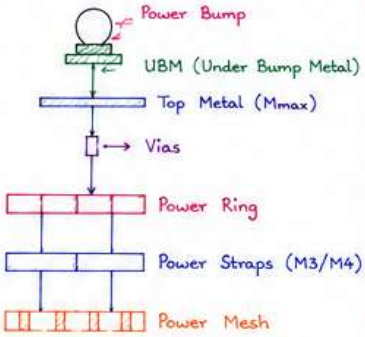
(d) Disadvantages

- ✗ Higher cost (bump & package cost).
- ✗ Requires flip-chip technology.
- ✗ Bump pitch limitation.
- ✗ Thermal management more critical.

Power Bump Types

Type	Description	Usage
VDD Bump	Provides V_{DD} to chip	Core / IO power
VSS Bump	Provides Vss to chip	Core / IO ground
Analog Bump	Provides AV_{DD} / AGND	Analog power
Mixed Bump	Power + Signal	High speed interfaces

(e) Power Bump Connection to PDN



Best Practices

- Use multiple V_{DD}/VSS bumps.
- Alternate V_{DD} and VSS bumps.
- Maximize bumps near high current blocks.
- Keep short & wide connections.
- Verify IR Drop and EM after implementation.
- Follow foundry bump rules.

4. Summary Table - Pads vs Bumps (Detailed)

Feature	Power Pads	Power Bumps
Where used	Wire-bond packages	Flip-chip packages
Path to chip	Pad → Bond wire → Package pin → Board	Bump → Substrate → Package
Current capability	Up to few hundred mA per pad	Several A per bump
Inductance	High (nH range)	Very low (pH range)
ESR	High	Low
Die size impact	Consumes pad area	No die edge consumption
Thermal performance	Moderate	Better
Use cases	Low/Medium perf. chips	High perf. SoC, CPU, GPU
Cost	Low	High
Reliability	Good	Very Good

5. Practical Observation

- × Poor pad/bump placement → IR Drop ↑, EM failure, timing issues.
- × Alternate V_{DD}/Vss reduces loop inductance.
- × Always connect pads/bumps to power rings using multiple vias.
- × Early planning of pads/bump count is critical.
- × Verify power integrity after each iteration.

6. Viva / Interview Points

Q1. What is the purpose of power pads ?
→ To bring power (V_{DD}) and ground (Vss) from package/board to on-chip PDN.

Q2. Why use power bumps instead of pads ?
→ Power bumps provide low inductance, high current capability and better power integrity.

Q3. How to reduce inductance in PDN interface ?
→ Alternate V_{DD}/Vss, short connections, multiple vias, use bumps.

7. Key Takeaways

- ✓ Pads are used in wire-bond packages.
- ✓ Bumps are used in flip-chip packages.
- ✓ Bumps provide low inductance & higher current.
- ✓ Proper placement reduces IR Drop and EM risk.
- ✓ Always follow foundry guidelines for pads and bumps.

8. Important Metrics

- IR Drop (mV / %)
- Current Density (mA/μm^2)
- EM Check (MTTF)
- Power Integrity (PI) Score
- Voltage Noise (mV)
- Decap Effectiveness (%)
- Power Efficiency (%)

Author: Konda Tharun Vivekananda

PD Power Planning

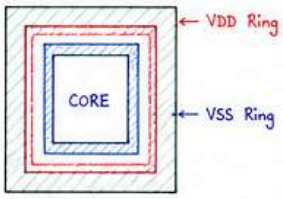
1. Power Rings

(a) Definition

Power Rings are closed metal loops placed around the core or macro to distribute power (VDD) and ground (VSS) uniformly and provide a low resistance path for high current.

(b) Functions

- Carry large current around the core.
- Reduce IR Drop and improve EM reliability.
- Provide low impedance connection to power straps.
- Protect core from voltage droop during high activity.
- Act as boundary for core area in floorplan.



(c) Advantages

- ✓ Low resistance path for bulk current.
- ✓ Ensures uniform voltage distribution.
- ✓ Improves IR Drop, EM and reliability.
- ✓ Supports connection to straps/mesh.
- ✓ Acts as shield and enhances signal integrity.

(d) Disadvantages

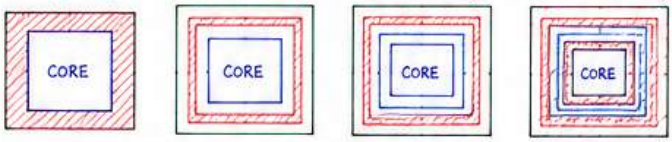
- ✗ Consumes routing resources & area.
- ✗ Large rings increase layout blockage.
- ✗ Improper sizing → IR Drop issues.
- ✗ Extra via cost to connect straps.
- ✗ Not effective alone → need straps/mesh.

Key Point

Rings handle bulk current along the periphery and act as highways, while straps/mesh distribute locally inside the core.

2. Types of Power Rings

(i) Single Ring (ii) Double Ring (iii) Quad Ring (iv) Nested Ring



VDD Ring VSS Ring

Type	Description	Use Case
Single Ring	One VDD + one VSS ring	Small blocks, low current
Double Ring	Two VDD + two VSS rings	Medium current designs
Quad Ring	Four VDD + four VSS rings	High current designs
Nested Ring	Rings at multiple offsets.	Large SoCs, better EM/IR

3. Ring Placement Guidelines

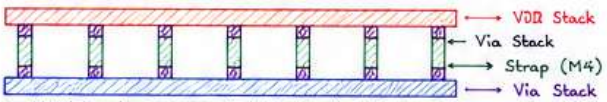
- Place rings on higher metal layers (M5/M6 or above).
- Keep rings close to core boundary but outside std. cell rows.
- Maintain clearance for DRC, density and antenna rules.
- Use recommended ring width & spacing from foundry.
- Ensure continuity with 45°/90° corners as per rule.
- Provide multiple vias to connect straps and mesh.



4. Ring Sizing Guidelines

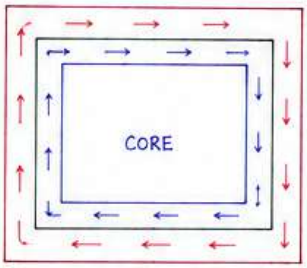
Parameter	Typical Range	Notes
Metal Layer	M5 – M8	Higher layer for low R
Ring Width (W)	5μm – 50μm	Depends on current
Ring Spacing (S)	2μm – 10μm	Between VDD & VSS
Enclosure	≥ 2μm	Keep out from core cells
Via Pitch	0.5μm – 1μm	Connect to straps/mesh

5. Ring to Strap Connection



- Multiple vias are required to handle high current.
- Via density must follow foundry EM/IR guidelines.

6. Ring Current Flow and Benefit



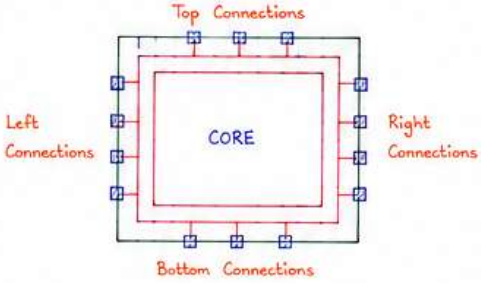
- Current from pads/bumps enters the rings.
- Rings carry current around the core.
- Current is distributed to straps/mesh via multiple vias.
- Low resistance loop → low IR Drop.

7. Rings vs Straps vs Mesh

Feature	Rings	Straps	Mesh
Location	Around core/macros	Inside core (vertical/horizontal)	Inside core (grid)
Handles	Bulk current	Column/row current	Local current
Direction	Closed loop	One direction	Both directions
Metal Layer	High (M5+)	Mid (M3–M6)	Low–Mid (M1–M4)
IR Drop	Low (globally)	Medium	Low (locally)
Area Overhead	Moderate	Low–Moderate	High
Use	High current path	Connect rings to mesh	Uniform distribution

8. Ring Connections

- Rings must connect to pads/bumps through straps.
- Provide at least 2 connections per side (more for large designs).
- Opposite sides should have equal number of connections.
- Avoid long unconnected ring segments.
- Ring break is not allowed → Maintain full continuity.



Recommended Min. Connections / Side

- Small IP : 2
- Medium Block : 4
- Large SoC : 4 to 8
- Very Large : 8+

9. Ring Design Checklist

- ✓ Ring metal width as per current requirement.
- ✓ Proper spacing between VDD & VSS rings.
- ✓ All corners follow 45° rule.
- ✓ Multiple vias to straps/mesh.
- ✓ Sufficient connections to pads/bumps.
- ✓ No ring breaks or floating segments.
- ✓ Density & DRC rules satisfied.
- ✓ IR Drop & EM checked after routing.
- ✓ Works with straps & mesh.
- ✓ Sign-off clean (IR, EM, DRC).

Practical Observation

- Wider rings reduce IR Drop but increase area.
- Too many rings may cause congestion near core boundary.
- Proper ring planning early (at floorplan stage) reduces ECOs.
- Ring effectiveness depends on good strap & mesh connectivity.
- Always validate using IR Drop & EM analysis after routing.

Key Note

Rings provide a low resistance closed loop for bulk power. They must be used with straps and mesh for complete PDN.

11. Interview / Viva Points

Q1. What are power rings ?
→ Closed metal loops around core to carry bulk power (VDD/VSS).

Q2. Why use power rings ?
→ To reduce IR Drop, EM issues and to provide low resistance path.

Q3. On which metal layers rings are placed ?
→ Higher metal layers (usually M5 and above).

12. Common Mistakes

- ✗ Using narrow ring width.
- ✗ Large spacing between VDD & VSS rings.
- ✗ Too few connections to straps/pads.
- ✗ Ignoring 45° corners and DRC.
- ✗ Ring breaks / discontinuity.
- ✗ Not verifying IR Drop & EM.

13. Important Metrics

- IR Drop (mV / %)
- Current Density (mA/μm²)
- EM Check (MTTF)
- Voltage Noise (mV)
- Power Integrity (PI) Score
- Reliability (EMIR Sign-off)

14. Summary Points

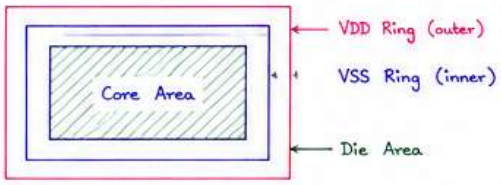
- Power rings carry bulk current around core.
- They reduce IR Drop and improve EM.
- Common types: Single, Double, Quad, Nested.
- Rings must connect to straps/mesh via multiple vias.
- Proper ring planning is essential for a robust PDN and successful tape-out.

Author: Konda Tharun Vivekananda

PD Power Planning

1. Power Rings – Structure

Power rings are closed loops of metal around the core/macro used to distribute power (VDD) and ground (VSS).
Usually implemented on higher metal layers (M5 or above).



2. Purpose / Why Power Rings ?

- Provide a low resistance, high current path around the core.
- Reduce IR Drop and improve EM reliability.
- Act as local power reservoirs during peak switching.
- Protect core from voltage drop during high activity.
- Provide connection points for straps, pads and macros.
- Improve signal integrity by stable power supply.

3. Functions

① Collects power from pads/bump/straps and distributes.
② Supplies uniform VDD / VSS to power straps & mesh.
③ Provides shielding effect against substrate noise.
④ Reduces loop area between supply and return paths.
⑤ Acts as backup power path in case of strap damage.

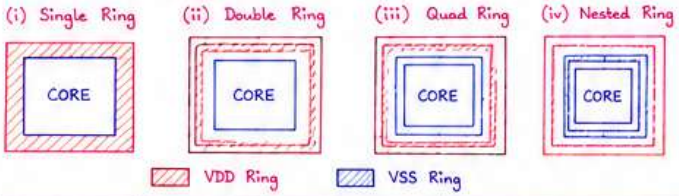
4. Advantages

✓ Low resistance path for very high current.
✓ Uniform voltage distribution across the core.
✓ Reduces IR Drop, EM & switching noise.
✓ Supports large designs and high performance SoCs.
✓ Improves reliability, stability and chip life.
✓ Easy to connect straps, pads, bumps & macros.

5. Disadvantages / Limitations

× Consumes routing resources and silicon area.
× Large rings increase die area / perimeter blockages.
× Additional cost for wide metal layers.
× Not effective alone → needs straps/mesh for distribution.
× Improper sizing → high IR Drop / EM failure.

6. Types of Power Rings

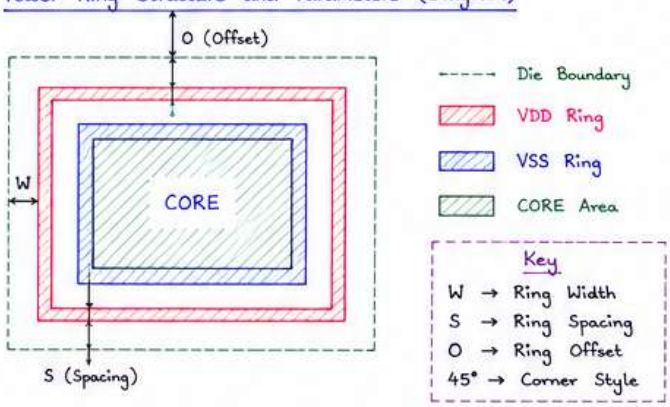


Type	Description	Use Case
Single Ring	One VDD + one VSS ring	Small blocks, low current
Double Ring	Two VDD + two VSS rings	Medium designs
Quad Ring	Four VDD + four VSS rings	High current designs
Nested Ring	Multiple alternate rings	Very large SoCs

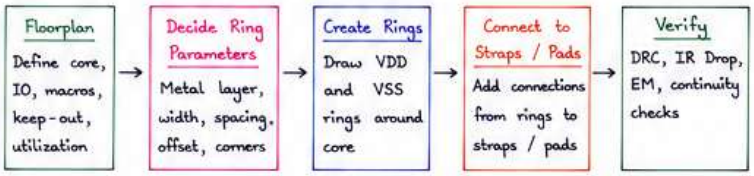
8. Power Ring Guidelines

- Place rings on higher metal layers (M5/M6 or above).
- Keep rings close to core boundary but outside std. cell rows.
- Maintain clearance for DRC, antenna and routing channels.
- Width should be decided based on current demand and IR Drop.
- Use even number of connections on all sides.
- Provide multiple connections to straps, pads and macros.
- Maintain 45° corners to satisfy DRC and EM rules.
- Follow foundry recommended rules for width, spacing, offset.
- Verify IR Drop and EM after implementation.

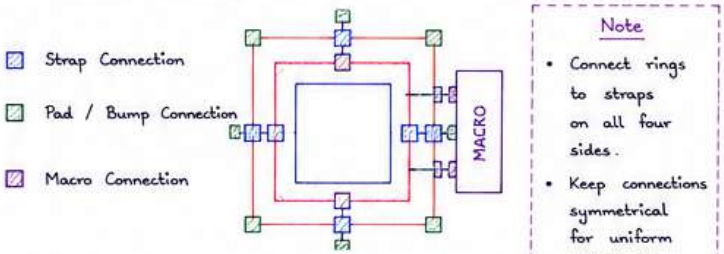
9. Power Ring Structure and Parameters (Diagram)



10. Power Ring Creation Flow (Steps)



11. Power Ring Connections (Typical)



12. Challenges with Power Rings

- High current in large designs → large ring width → area penalty.
- Improper ring sizing → IR Drop, EM failure.
- Too few connections → current crowding.
- Ring blocks routing channels and macro placement.
- In advanced nodes, high current density limits (EM) are strict.

13. Practical Observation

- Wider rings reduce IR Drop but increase area.
- Alternate VDD/VSS connections help balance current.
- Always validate with IR Drop and EM after routing.
- Good early planning saves ECOs and iterations.

14. Interview / Viva Points

Q1. What are power rings ?
→ Closed metal loops around core to distribute VDD/VSS.

Q2. Why alternate VDD and VSS ?
→ To reduce loop area, noise and IR Drop.

Q3. On which metal layers rings are usually placed ?
→ Higher metal layers (M5 or above).

15. Key Takeaways

✓ Power rings provide a low resistance path for power.
✓ Rings surround core/macro and connect to straps/pads.
✓ Proper sizing & connections reduce IR Drop & EM.
✓ Rings + straps + mesh together ensure robust PDN.

16. Important Metrics

- IR Drop (mV / %)
- Current Density (mA/μm²)
- EM Check (MTTF)
- Voltage Noise (mV)
- Power Integrity (PI) Score
- Decap Effectiveness (%)
- Power Efficiency (%)

17. Summary / Revision Points

1. Power rings are closed metal loops surrounding the core.
2. They carry bulk current and provide low resistance path.
3. Rings reduce IR Drop, improve EM reliability and stability.
4. Common types: Single, Double, Quad and Nested rings.
5. Rings are placed on higher metal layers and follow DRC rules.
6. Width, spacings, offset and connections are key parameters.
7. Rings connect to straps, pads and bumps for distribution.
8. Proper planning reduces noise, improves SI and reliability.
9. Rings alone are not enough → need straps and mesh.
10. Always verify IR Drop, EM and continuity after implementation.

PD Power Planning

1. Power Straps – Introduction

Power Straps are wide metal lines that run across the core area to distribute power (VDD) and ground (VSS) from Power Rings to the Power Mesh. They carry bulk current and reduce IR Drop and EM issues.

2. Purpose / Why Straps ?

- Carry large current across the core.
- Distribute power uniformly to the mesh.
- Reduce IR Drop and EM violations.
- Provide low impedance path for switching activity.
- Connect rings to mesh via multiple vias.
- Improve timing, reliability and silicon integrity.

3. Functions

① Collect current from rings.
② Transport high current across long distances.
③ Feed current into the mesh through vias.
④ Balance power density across the core.
⑤ Act as backbone for PDN.

4. Advantages

✓ Low resistance path for bulk current.
✓ Uniform voltage distribution across core.
✓ Reduces IR Drop, EM and voltage noise.
✓ Supports high performance and large SoCs.
✓ Easy to connect to rings/mesh with multiple vias.
✓ Improves reliability and lifetime.

5. Challenges / Limitations

× Too wide straps consume routing tracks and area.
× Improper spacing → IR Drop and EM issues.
× Straps block routing channels → congestion.
× Need careful ring to strap connections.
× Additional vias increase capacitance and cost.

6. Types of Power Straps (9 Types)

1. Horizontal Strap — Runs left to right	2. Vertical Strap — Runs bottom to top	3. Double Horizontal Strap — Two sets, left to right
4. Double Vertical Strap — Two sets, bottom to top	5. Grid Strap — Horizontal + Vertical	6. Staggered Strap — Offset between layers
7. Wide Strap — Higher width strap	8. Shield Strap — On boundaries / shield	9. Offset Strap — At specific offset

VDD Strap VSS Strap

7. Strap Parameters

Parameter	Description	Typical Range / Notes
Layer (Metal)	Metal layer used	M4 – M8 or above
Orientation	Horizontal / Vertical	Both generally used
Width (W)	Width of strap	2 µm – 50 µm (based on current)
Spacing (S)	Spacing between straps	2 µm – 20 µm (min DRC)
Pitch (P)	W + S	Depends on current density
Via Size	Via connecting to mesh	Min via / Stack via
Via Pitch	Distance between vias	2 × via pitch rule
Offset	Distance from die edge	2 µm – 20 µm
Current Density (J)	Current per unit area	0.8 – 1.5 mA/µm² (Cu)

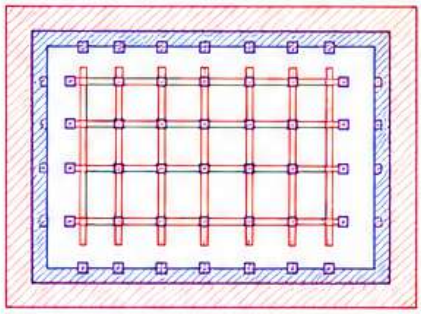
8. Strap Guidelines

- Place straps on higher metal layers (M4 or above).
- Use both horizontal and vertical straps.
- Keep strap width based on current demand.
- Maintain even spacing and pitch.
- Connect straps to rings on both sides.
- Provide multiple vias to mesh at regular interval.
- Keep straps away from standard cell rows by min offset.
- Follow 45° corner rule and DRC guidelines.
- Verify IR Drop and EM after implementation.

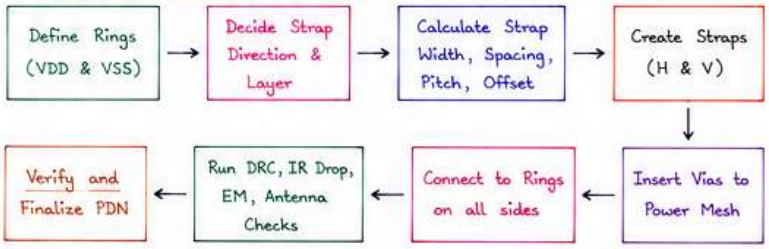
Key Point

Straps act as highways for current, transporting high current from rings to the mesh with low resistance.

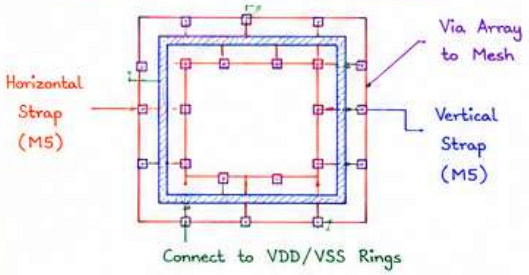
9. Strap Structure (Example)



10. Strap Creation Flow (Steps)



11. Strap Connection to Rings and Mesh



Recommended Connections

- Min 2 Connections per side
- Via pitch as per foundry rules
- Symmetrical connections

12. Strap vs Ring vs Mesh – Role

Component	Role	Current Handled	Location
Power Rings	Collect from pads/bumps, distribute around core	Very High (Bulk)	Around core / Macros
Power Straps	Transport current across core to mesh	High	Inside core (Grid lines)
Power Mesh	Distribute power to cells locally	Low – Medium	Inside core (All over)

Practical Observation

- Too few straps → IR Drop ↑, timing failures.
- Too many straps → routing blockage, area ↑.
- Proper strap pitch & width is critical.
- Always check IR Drop/EM after route.
- Early planning of straps reduces redesign.

14. Important Metrics

- IR Drop (mV / %)
- Current Density (mA/µm²)
- EM Check (MTTF)
- Power Integrity (PI) Score
- Voltage Noise (mV)
- Decap Effectiveness (%)
- Power Efficiency (%)

15. Common Mistakes

× Wrong strap direction / imbalance.
× Wrong strap width (too small).
× Irregular spacing and pitch.
× Not connecting straps to rings.
× Missing vias to mesh.
× Not checking IR Drop / EM early.

16. Viva / Interview Points

Q1. What are power straps ?
→ Wide metal lines used to carry bulk current across core.

Q2. Why both H and V straps used ?
→ To reduce path length and distribute current uniformly.

Q3. How straps connect to mesh ?
→ Through multiple vias at regular pitch.

17. Key Takeaways

✓ Straps carry bulk current from rings to mesh.
✓ Proper width, spacing and connections ensure low IR Drop and EM.
✓ Use grid of H & V straps for uniform distribution.
✓ Early PD planning avoids congestion and timing issues.

18. Summary / Revision Points

1. Straps are wide metal lines for power distribution.
2. Types: 9 major strap patterns.
3. Key parameters: width, spacing, pitch, layer, via pitch.
4. Connect straps to rings on all sides.
5. Provide multiple vias to mesh.
6. Follow guidelines and DRC rules.
7. Verify IR Drop and EM after implementation.
8. Good strap planning → robust PDN.

Author: Konda Tharun Vivekananda

PD Power Planning

1. Power Vias & Via Arrays - Introduction

- Vias provide vertical electrical connection between two metal layers.
- Via Arrays are group of vias placed together to carry large current and reduce resistance.
- Used in PDN to connect: Straps ↔ Straps, Straps ↔ Rings, Rings ↔ Pads/Bumps, Straps ↔ Mesh, etc.
- Helps to reduce IR Drop, EM issues and improves reliability.

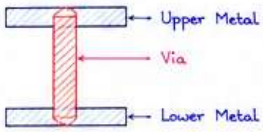
2. Purpose / Why Via Arrays ?

- Carry high current between metal layers safely.
- Provide multiple parallel paths → reduce resistance.
- Reduce IR Drop and improve EM reliability.
- Ensure uniform power distribution across PDN.
- Support connections at critical locations (rings, straps, pads).
- Improve signal integrity by stable power supply.

3. Functions

① Connect different metal layers in PDN.
② Provide low resistance vertical path.
③ Distribute current uniformly.
④ Provide redundancy and reliability.
⑤ Reduce current density per via.
⑥ Act as bridge between rings, straps, pads and mesh.

Via Cross Section



4. Types of Vias

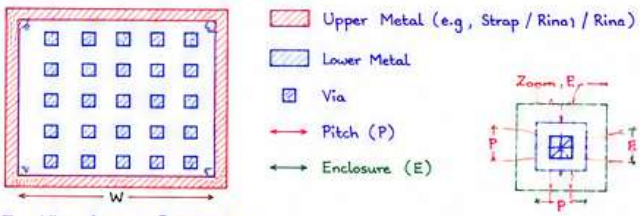
Type	Description	Typical Layers
Contact	Connects Metal1 to Active / Poly	M1 ↔ Active / Poly
Via 1	Connects Metal1 to Metal2	M1 ↔ M2
Via 2	Connects Metal2 to Metal3	M2 ↔ M3
Via 3	Connects Metal3 to Metal4	M3 ↔ M4
Via n	Connects Metal n to Metal n+1	Mn ↔ Mn+1

5. Types of Via Arrays

1. Single Via — One via only	2. Line (1D) Array — Vias in single line	3. Row-Column Array — Rectangular (rows x cols)
4. Staggered Array — Offset every alternate row	5. Diagonal Array — Vias along diagonal	6. Enclosed Array — Vias around perimeter
7. Cross (Plus) Array — Plus shape connection	8. Dense Matrix Array — Dense matrix for high current	9. Custom (Asymmetric) — Based on routing need

(All arrays can be of any size depending on current requirement)

6. Via Array Structure (Example)



7. Via Array Parameters

Parameter	Symbol	Description	Notes / Typical
Via Pitch	P	Center to center distance between vias	Min by DRC
Via Enclosure	E	Metal overhang around via	Min by DRC
Array Width	W	Overall width of array	Depends on current
Array Height	H	Overall height of array	Depends on current
No. of Vias (X)	Nx	Columns in array	>= 1
No. of Vias (Y)	Ny	Rows in array	>= 1
Current Capacity	Iarray	Total current carried	= n x Ivia (approx)

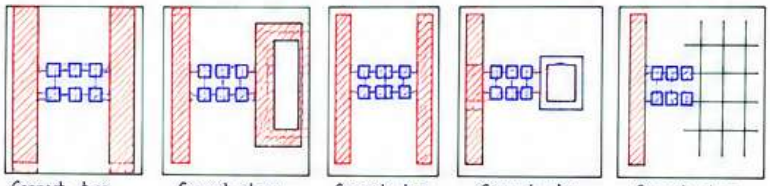
Current Capacity ≈ Iarray = Nx × Ny × Ivia × Kutil
where Ivia = current per via, Kutil = utilization factor (0.6 ~ 0.8)

8. Via Array Guidelines

- Use as many vias as needed to meet current requirement.
- Follow DRC rules for pitch, enclosure and spacing.
- Keep via array close to connection point (ring/strap/pad).
- Symmetrical arrays are preferred for uniform current flow.
- Avoid long chains of single vias → use arrays instead.
- Leave proper space for routing and manufacturing.
- Verify IR Drop and EM after implementation.

Key Point
More vias in parallel ↓ Resistance & IR Drop
↑ Current Handling

9. Via Array Placement Examples



10. Via Array vs Single Via - Comparison

Feature	Single Via	Via Array
Current Capacity	Low	Very High
Resistance	High	Low (parallel paths)
IR Drop	High	Low
EM Reliability	Low	High
Redundancy	None	High
Area Overhead	Low	Medium / High
Use Case	Low current / signals	Power distribution (PDN)

11. Advantages

- ✓ Very high current handling capability.
- ✓ Low resistance path.
- ✓ Improves IR Drop and EM reliability.
- ✓ Provides redundancy.
- ✓ Better PDN stability and reliability.
- ✓ Flexible sizing based on current need.

12. Challenges

- × More area overhead.
- × Increase routing congestion.
- × Too many vias → DRC issues.
- × Improper placement → IR Drop still high.
- × Additional manufacturing cost.

Practical Observation

- Early planning of via arrays avoids ECOs.
- Place large via arrays near high current entry/exit points.
- Always check electromigration (EM) after routing.
- Use staggered or row-column arrays for better current sharing.
- Over design is safe but increases area and congestion.

Good Practice

- Verify PDN with IR Drop and EM.
- Follow DRC & Foundry rules.
- Maintain symmetry for uniform current.

14. Via Array Creation Flow (Steps)

Identify Connection Points (Rings/Straps/Pads/Mesh) → Decide Via Layer Pair (e.g., M4-M5) → Calculate Required Vias (Current Based) → Place Via Array (Pitch, Enclosure, Rows x Cols) → Run DRC, EM Checks → Finalize PDN

15. Recommended Min. Values (Typical)

Layer Pair	Pitch (P)	Enclosure (E)	Via Size	Notes
M1-M2	0.28 - 0.40 µm	0.08 - 0.10 µm	0.10 - 0.12 µm	Depends on tech
M2-M3	0.30 - 0.45 µm	0.08 - 0.10 µm	0.12 - 0.14 µm	Depends on tech
M3-M4	0.36 - 0.50 µm	0.10 - 0.12 µm	0.14 - 0.16 µm	Depends on tech
M4-M5	0.40 - 0.60 µm	0.12 - 0.14 µm	0.16 - 0.18 µm	PDN critical

Note: Always refer Foundry DRC manual for exact values.

16. Interview / Viva Points

Q1. What is the purpose of via arrays ?
→ To carry high current between layers with low resistance and low IR Drop.
Q2. Why are via arrays preferred over single vias in PDN ?
→ Because they increase current capacity and reduce EM & IR Drop.
Q3. What are key parametres of via array ?

17. Common Mistakes

- × Using single via for high current.
- × Too small pitch or enclosure → DRC violations.
- × Overlooking EM checks.
- × Long distance between connection points and via arrays.
- × Asymmetric or uneven via arrays.

18. Important Metrics

- IR Drop (mV / %)
- Current Density (mA/µm²)
- EM Check (MTTF)
- Power Integrity (PI) Score
- Voltage Noise (mV)
- Power Efficiency (%)

19. Key Takeaways

- ✓ Via arrays are essential for high current PDN connections.
- ✓ Proper sizing and placement ensure low IR Drop and EM safe.
- ✓ Follow foundry guidelines for pitch, enclosure and rules.

20. Summary Points

1. Vias connect different metal layers.
2. Via arrays provide parallel current paths.
3. Reduce resistance, IR Drop and EM.
4. Types include line, row-column, staggered, cross, dense matrix etc.
5. Follow DRC, IR Drop and EM rules.
6. Proper planning → robust PDN.

Author: Konda Tharun Vivekananda

PD Power Planning

1. Power Vias & Via Arrays - Introduction

- Power Vias connect two adjacent metal layers vertically.
- Via Arrays are a group of vias placed together to carry high current and reduce resistance.
- Used to connect: Power Rings ↔ Straps ↔ Pads/Bumps ↔ Mesh
- Proper via usage is critical to control IR Drop, EM and ensure robust PDN.

2. Purpose / Why Via Arrays?

- Provide low resistance vertical connections.
- Increase current carrying capability.
- Reduce current density per via.
- Improve reliability by reducing EM and IR Drop.
- Support flexible power distribution across the core.

3. Types of Power Vias

Type	Description	Typical Layers	Use Case
Contact	Metal1 to Active/Poly	M1 ↔ Active/Poly	Cell connections
Via 1	Metal1 to Metal2	M1 ↔ M2	Local cell power
Via 2	Metal2 to Metal3	M2 ↔ M3	Strap connections
Via 3	Metal3 to Metal4	M3 ↔ M4	High current path
Via 4	Metal4 to Metal5	M4 ↔ M5	Core ring to straps
Via 5	Metal5 to Metal6	M5 ↔ M6	Upper level straps
Via 6	Metal6 to Metal7	M6 ↔ M7	Long distance path
Via 7	Metal7 to Metal8	M7 ↔ M8	Top level mesh
Via 8+	Higher metal connections	Mn ↔ Mn+1	Advanced nodes

4. Types of Via Arrays (9 Types)

1. Single Via — One via only	2. Line (1D) Array — Vias in single line	3. Row-Column Array — Rectangular rows x cols
4. Staggered Array — Offset every alternate row	5. Diagonal Array — Vias along diagonal	6. Enclosed (Perimeter) Array — Vias around perimeter
7. Cross (Plus) Array — Plus shape connection	8. Dense Matrix Array — Dense matrix for high current	9. Custom (Asymmetric) Array — Based on routing / current need

5. Via Array Parameters

Parameter	Symbol	Description	Typical / Notes
Via Pitch (both X & Y)	P	Center to center distance	0.2 – 0.6 µm (tech dep.)
Enclosure (E)	E	Metal overlap around via	>= Min by DRC
Array Rows	R	Number of rows in array	>= 1
Array Columns	C	Number of columns in array	>= 1
Array Width (W)	W	Overall width ((C-1)*P + 2*E)	Depends on current
Array Height (H)	H	Overall height ((R-1)*P + 2*E)	Depends on current
Current Capacity (I)	I_{array}	Total current carried	≈ N_{vias} × I_{via} × K_{util}
Utilization Factor	K_{util}	Via current derating factor	0.6 – 0.8 (typ.)

Note: Always follow foundry PDK for min pitch, enclosure, cut rules.

6. Functions

1. Carry high current between power structures.
2. Reduce vertical resistance and voltage drop.
3. Distribute current uniformly to upper/lower layers.
4. Provide backup paths in case of single via failure.
5. Improve EM reliability by lowering current density.

7. Advantages

- ✓ Higher current handling capability.
- ✓ Lower IR Drop and EM risk.
- ✓ Redundant and reliable connections.
- ✓ Improves PDN stability & lifetime.
- ✓ Supports advanced nodes & high perf. SoCs.

8. Challenges / Limitations

- X Requires more area (especially large arrays).
- X Improper use → routing congestion.
- X Too large arrays increase capacitive coupling.
- X EM still possible if arrays are insufficient.
- X Foundry DRC/EM rules are strict in advanced nodes.

9. Via Array Guidelines

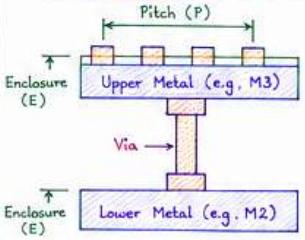
- Use arrays for high current paths, not single vias.
- Keep pitch ≥ min pitch from foundry.
- Maintain required enclosure on all sides.
- Prefer symmetrical arrays for uniform current.
- Do not place vias on metal corners (use 45° rule).
- Avoid long unconnected via chains.
- Use odd number of rows/columns for better symmetry.
- Follow EM and DRC rules strictly.

Key Point

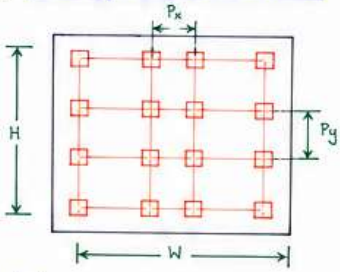
More Via in Parallel ↓ Resistance & IR Drop

↑ Current Handling

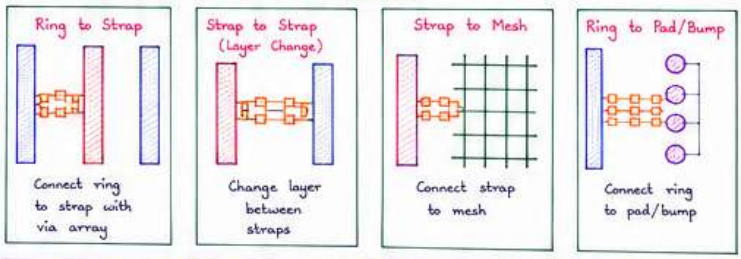
10. Via Structure (Cross Section)



11. Via Array Example (Row-Column)



12. Via Array Placement in PDN (Examples)



Aspect	Single Via	Via Array	Role in PDN
Current Capability	Low	High	Carry bulk current
Resistance	High	Low	Reduce resistance
IR Drop	High	Low	Improve voltage
EM Reliability	Low	High	Better EM margin
Area	Very small	Larger	Trade-off with area
Use Case	Low current nets	Power nets	Robust power path

14. Practical Observation

- Big arrays reduce IR Drop but need more area.
- Staggered / cross arrays help in congestion reduction.
- Always validate with IR Drop & EM analysis.
- Poor via planning → hot spots, performance loss.
- Early planning of via locations saves ECOs later.

Good Practice

- Use via arrays at all critical connections.
- Maintain symmetry and uniformity.
- Follow DRC & foundry rules.

14. Pena Vipak Observation

1. What is the difference between via and via array?
→ Via is a single connection between two layers.
→ Via array is a group of vias to carry high current.

1. Yink tialon pie for high current peations.
3. Austit knocle otoap and raunees daft ooke.
3. Sink tina chaliiy is cunving dethgs.

15. Complete Power Plan Flow (Stages Explained)

1. **Floorplan Read** — Read floorplan, macros, IO, tech, constraints. Understand chip structure.
2. **Power Estimation** — Estimate total power, peak current, activity per domain.
3. **Voltage Domain Planning** — Define power domains and voltages. Plan isolation and level shifting.
4. **Pad / Bump Planning** — Plan location and count of pads/bumps for VDD/VSS.
5. **Core Ring** — Create closed rings around core for low R return path.
6. **Macro Ring** — Create rings around macros to isolate and supply power.
7. **Power Straps** — Add vertical & horizontal straps to carry bulk current.
8. **Power Mesh** — Create mesh on higher layers for uniform distribution.
9. **Via Reinforcement** — Add multiple via arrays at critical connections.
10. **Decap Placement** — Place decaps near high switching regions.
11. **Std-Cell Power Rails** — Ensure std-cell rails connectivity to straps/mesh via vias.
12. **Special Power Structures** — Add special structures like deep N-well, dvfs cells, etc.
13. **Power Domain Connectivity** — Ensure proper connection between all power domains.
14. **Connectivity Verification** — Check opens, shorts and correct connectivity.
15. **IR Drop Analysis** — Analyze voltage drop across PDN, identify hot spots.
16. **EM Analysis Fix Optimization** — Check current density and electromigration limits.
18. **EMIR Sign-Off** — Final sign-off for IR Drop, EM, reliability and SI.

16. Viva / Interview Points

Q1. What is the difference between via and via array?
→ Via is a single connection between two layers.
→ Via array is a group of vias to carry high current.
Q2. Why we use via arrays in PDN?
→ To reduce resistance, IR Drop and EM issues.
Q3. What decides the size of via array?
→ Current requirement, layer pair, DRC rules and EM limits.

Common Mistakes

- X Using single via for high current path.
- X Pitch less than recommended.
- X Missing enclosure or corner rule.
- X Asymmetric arrays causing imbalance.
- X Not checking IR Drop / EM after route.

18. Key Takeaways

- ✓ Via arrays are essential for robust PDN.
- ✓ More vias in parallel → less R & IR Drop.
- ✓ Proper pitch, enclosure & symmetry ensure reliability.
- ✓ Verify with EMIR for safe design.

19. Important Metrics

- IR Drop (mV / %)
- Current Density (mA/µm²)
- EM Check (MTTF)
- Voltage Noise (mV)
- Power Integrity (PI) Score
- Decap Effectiveness (%)
- Power Efficiency (%)

Author: Konda Tharun Vivekananda

PD Power Planning

1. Voltage Domain - Definition

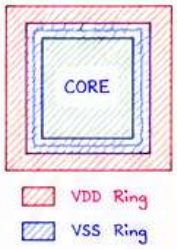
- A voltage Domain is a group of logic/circuits that operate at the same supply voltage (VDD) and ground (VSS).
- Each domain has its own power intent and power network.
- Multiple voltage domains are used in low power SoCs to support different performance and power requirements.

Reasons / Why Voltage Domains ?

- To reduce power by using lower voltage in non-critical blocks.
- To isolate noisy or high switching blocks.
- To allow independent power shut-down.
- To meet multi-mode, multi-performance requirements.
- Improve reliability by limiting IR Drop impact.

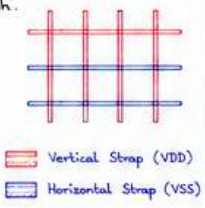
2. Benefits of Power Rings

- ✓ Provide a low resistance closed loop around core/macros.
- ✓ Act as local power reservoirs during peak current.
- ✓ Reduce IR Drop and improve EM reliability.
- Provide connection points for straps, pads and macros.
- Improve signal integrity by stable supply.
- Help in ESD robustness and noise isolation.
- Ease of routing and power distribution planning.

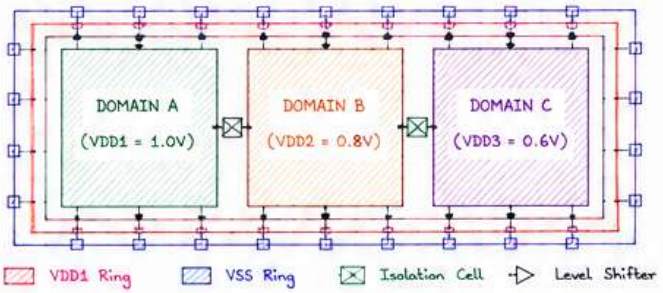


3. Benefits of Power Straps

- ✓ Transport high current from rings to the power mesh.
- ✓ Distribute power uniformly across the core.
- ✓ Reduce IR Drop over long distances.
- ✓ Provide multiple low impedance paths.
- ✓ Balance current density across layers.
- ✓ Improve timing, SI and reliability.
- ✓ Enhance robustness for large and complex designs.



4. Typical Voltage Domain Example



5. Comparison - Single Domain vs Multi-Domain

Feature	Single Voltage Domain	Multi Voltage Domains
Power	High (all at high V)	Lower (reduce V in non-critical)
Performance	Uniform high perf.	Optimized per block.
Design Complexity	Low	High (isolation + shifters)
IR Drop Impact	High	Lower (smaller currents)
Area	Less	More (extra cells)
Use Case	Small / simple designs	Large SoCs, Low Power

6. Practical Observation

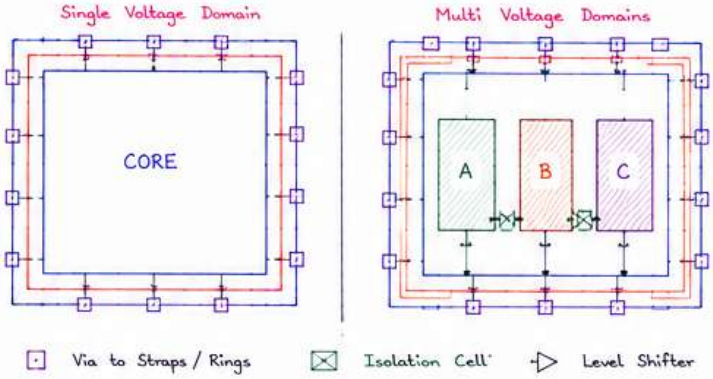
- Proper domain planning reduces leakage & dynamic power significantly.
- Always validate isolation, level shifting and power intent.
- Use ring + strap + mesh together for robust PDN.

7. Challenges

- x Complex power intent and planning.
- x IR Drop at domain boundaries.
- x Too few straps and isolation overhead.
- x Power gating wake-up latency.
- x EM issues if domains are imbalanced.

8. Voltage Domains in PDN - Key Points

- Each domain has separate power rings and connections.
- Domains may share VSS but have different VDD levels.
- Always check domain boundary crossings.
- Ensure correct power intent (UPF/CPF).
- Verify IR Drop, EM and continuity per domain.



9. Power Rings & Straps - Role in PDN

Structure	Role
Power Rings	• Collect power from pads/bumps. • Act as local reservoir. • Provide low resistance loop.
Power Straps	• Carry high current to the core. • Connect rings to mesh. • Distribute power uniformly.

10. Design Guidelines for Voltage Domains

- Group blocks with similar perf & power into same domain.
- Keep high switching blocks in separate domain.
- Use lowest possible voltage for non-critical blocks.
- Place isolation cells on power gated domain boundaries.
- Follow foundry rules for level shifters and clamp cells.
- Verify power intent early in the flow.

11. Power Rings & Straps Parameters

Parameter	Symbol	Description	Typical / Notes
Ring Width	W_{ring}	Width of ring metal	5 - 20 µm (typ.)
Ring Spacing	S_{ring}	Spacing between VDD & VSS rings	2 - 10 µm
Strap Width	W_{strap}	Width of straps	2 - 10 µm
Strap Pitch	P_{strap}	Distance between straps	10 - 50 µm
Via Size	V_{via}	Via cut size	Depends on layer
Via Pitch	P_{via}	Pitch between vias	2 - 10 µm
Current Density	J	Current per area	< 1 - 2 mA/µm²

12. Summary / Key Takeaways

- ✓ Voltage Domains help in low power and performance optimization.
- ✓ Power Rings provide low resistance loop and local reservoirs.
- ✓ Power Straps transport high current from rings to mesh.
- ✓ Together Rings + Straps + Mesh form a robust PDN.
- ✓ Proper domain planning, IR/EM analysis and verification ensure reliable and efficient power distribution.

13. Interview / Viva Points

Q1. What is a voltage domain ?
→ Set of logic blocks operating at same supply voltage (VDD) and ground (VSS).

Q2. Why use multiple voltage domains ?
→ To reduce power, isolate noise and support multi-performance.

Q3. What is the role of power rings ?
→ Provide low resistance loop, reservoir and connection points.

14. Common Mistakes

- x Incorrect domain grouping.
- x Missing isolation / level shifters.
- x Too few straps causing IR Drop.
- x Ignoring EM rules.
- x Not verifying power intent.

15. Important Metrics

- IR Drop (mV / %)
- Current Density (mA/µm²)
- EM Check (MTTF)
- Power Integrity (PI) Score
- Decap Effectiveness (%)
- Power Efficiency (%)

16. Signs of Poor Power Planning

- x High IR Drop and voltage droop.
- x Timing failures and noise issues.
- x EM violations in metals/vias.
- x Power mesh congestion.
- x Instability and reliability failures.

Author: Konda Tharun Vivekananda

PD Power Planning

Power Connectivity Verification

1. Introduction

- Power Connectivity Verification (PCV) ensures that all power (VDD) and ground (VSS) connections in the design are complete, correct and as per intent.
- It checks short, open, missing, floating and wrong connections in the entire power distribution network (PDN).
- PCV is a sign-off quality check before IR/EM analysis and tape-out.

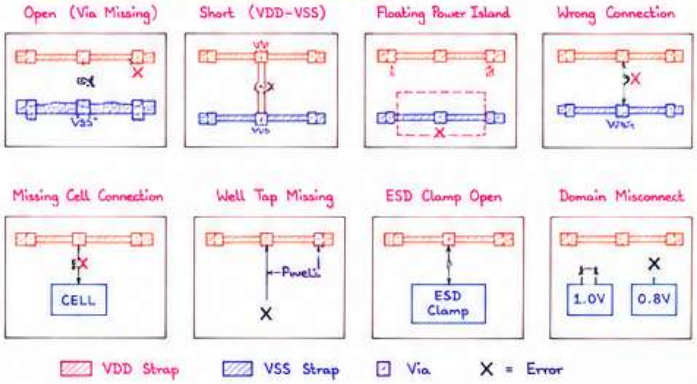
2. Purpose / Why PCV ?

- To ensure every cell, macro and pad gets proper power and ground.
- To detect opens, shorts, missing vias, wrong connections early.
- To avoid floating power islands and unintended power paths.
- To verify ESD, clamp, well and substrate connections.
- To maintain reliability, prevent EM/IR violations and silicon failure.

3. What is Checked in PCV ?

① Opens : Missing connections in rings, straps, vias.
② Shorts : VDD to VSS or between different voltage domains.
③ Missing Connections : Cells/macros not connected to PDN.
④ Floating Nets : Power/ground nets not tied to any source.
⑤ Wrong Connections : VDD ⟷ VSS swapped, domain misconnections.
⑥ Well/Substrate : Pwell / Nwell and substrate taps connectivity.
⑦ ESD Structures : VDD/VSS to ESD rings and clamps.
⑧ Special Nets : Analog power pins, bias nets connectivity.

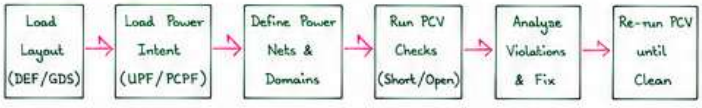
4. Common PCV Error Types (Examples)



5. PCV Checks by Category

Category	Checks Performed	Why Important
Global PDN	Ring, straps, mesh connectivity	Ensure complete power grid
Local Connection	Via stacks, via arrays, tap cells	Avoid opens & high resistance
Cell/Macro	Power pins connected	Prevent floating elements
Power Domains	Isolation, level shifter, domain links	Avoid short & cross-domain issues
Well/Substrate	Pwell/Nwell ties, substrate taps	Prevent latch-up & reliability issues
ESD Network	Clamp to rails, ESD ring ties	ESD robustness & compliance
Special Nets	Analog/Bias power pins	Functional correctness

6. PCV Typical Flow



Key Point

PCV ensures that the designed Power Grid is correctly built and connected as per intent before IR/EM sign-off.

Practical Observation

- Catching opens and missing vias early saves time.
- Most IR Drop issues root cause is floating connectivity.
- Always run PCV after any ECO or power grid change.
- Use via arrays at high current points to avoid opens.

7. PCV Algorithms (How it Works)

- Extract power connectivity graph from layout.
- Compare against power intent (UPF / PCPF).
- Traverse each power net & check connectivity to sources.
- Identify disconnected components, shorts, floating nets.
- Report violations with location, layer and net information.
- Use DRC rules for via stacks and enclosure verification.
- Support hierarchical and flat verification.

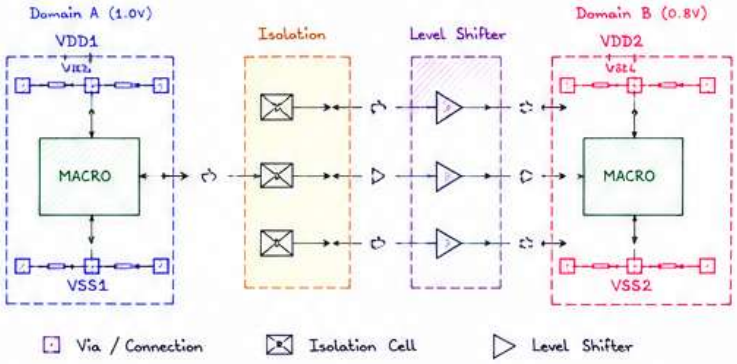
Remember

Good layout is not enough, Good connectivity is must.

8. PCV Results Classification

Result Type	Description
Clean	No opens, shorts, missing or floating nets.
Critical	Power shorts, open to source, missing power to macro/cell.
High	High resistance connections, long via chains missing.
Medium	Sub-optimal connections, weak domain links.
Low	Minor issues, non-critical warnings.

9. Power Domain Connectivity Check (Example)



10. Sign-off Criteria (PCV)

- ✓ No power shorts between VDD and VSS.
- ✓ No opens in rings, straps, mesh and via stacks.
- ✓ All cells, macros and pads connected to correct domain.
- ✓ No floating power or ground nets.
- ✓ Well/substrate and ESD connections complete.
- ✓ No cross connection between different voltage domains.
- ✓ All violations classified as Clean before IR/EM analysis.

PCV is a Mandatory Step before Power Integrity Sign-off.

11. Benefits of Good Power Connectivity

- ✓ Reliable and robust power distribution.
- ✓ Lower IR Drop and EM stress.
- ✓ Improved chip performance and stability.
- ✓ Reduced silicon failures and respins.
- ✓ Early detection of layout and design issues.

Good Connectivity = Good Silicon + Better Yield

12. Challenges in PCV

- ✗ Complex multi-domain SOCs with many supplies.
- ✗ Large via stacks and deep routing layers.
- ✗ ECO changes may break existing connections.
- ✗ Analog/mixed-signal macros with special pins.
- ✗ Advanced nodes with strict DRC on vias and enclosures.

12. Interview / Viva Points

Q1. What is PCV in PDN ?
→ It verifies all power/ground connections to ensure no opens, shorts or floating nets.

Q2. Why is PCV important ?
→ To avoid IR Drop, EM issues and silicon failures.

14. Summary / Revision Points

- PCV ensures correct connectivity of entire power distribution network.
- It checks opens, shorts, missing and floating connections.
- Verifies cells, macros, rings, straps, vias, wells and ESD structures.
- Uses power intent to validate domain and connectivity rules.
- Mandatory before IR Drop and EM analysis.
- Clean PCV results lead to reliable and robust silicon.

PD Power Planning

1. Static IR Drop – Definition

- Static IR Drop is the steady state voltage drop across the power distribution network (PDN) due to DC current.
- It occurs even when the circuit is not switching.
- Caused by the resistance of power rings, straps, mesh, vias and rails.
- If large, it reduces the effective supply voltage → performance loss or functional failure.

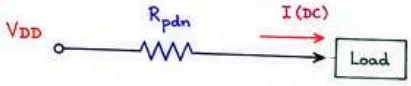
2. Static IR Drop – Factors Increasing

① High current demand (large leakage or always-on blocks).
② High resistance in the PDN (long / wide metal, thin metal).
③ Long distance from power source to the load.
④ Insufficient power rings / straps / vias.
⑤ Poor via utilization / single vias.
⑥ High metal resistivity (thin layer usage).
⑦ Large macros far from power rings.
⑧ High temperature (↑ R of metal).

3. Static IR Drop – Fixes / Mitigation

✓ Increase metal width and use lower metal layers.
✓ Add more power rings and multiple rings.
✓ Reduce distance between rings and straps.
✓ Increase number of straps and strap width.
✓ Add more via arrays and via reinforcement.
✓ Use multiple power rails for std cells.
✓ Add local decoupling capacitors (decaps).
✓ Balance current distribution using domain planning.

4. Static IR Drop – Illustration



$$IR\ Drop = I \times R_{pdn}$$
$$V_{load} = V_{DD} - (I \times R_{pdn})$$

I : DC Current
R_{pdn} : PDN Resistance
V_{load} : Voltage at Load

5. Dynamic IR Drop – Definition

- Dynamic IR Drop is the instantaneous voltage drop due to sudden switching activity and transient current.
- Occurs during signal transitions.
- Caused by the inductance (L) and resistance (R) of PDN.
- Leads to momentary voltage droop or overshoot.

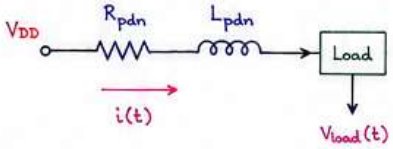
6. Dynamic IR Drop – Factors Increasing

① High switching activity (many simultaneous toggles).
② High di/dt (fast current change).
③ High PDN inductance (long metal / via, large loops).
④ Sparse decap capacitors.
⑤ Long unconnected via chains (stubs).
⑥ Weak power grid (small width / insufficient mesh).
⑦ Coupling between power and signal nets.
⑧ Advanced nodes (low VDD, high current density).

7. Dynamic IR Drop – Fixes / Mitigation

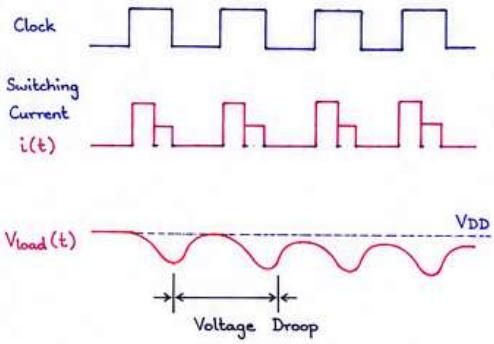
✓ Add decoupling capacitors near high switching blocks.
✓ Use multiple power straps close to each other.
✓ Increase planar se mesh density.
✓ Reduce loop inductance and via loop area.
✓ Add via arrays and reduce via chain length.
✓ Use lower metal layers for power distribution.
✓ Bale domains and power intent (UPF/CPF).
✓ Optimize placement to reduce simultaneous switching.

8. Dynamic IR Drop – Illustration



$$V_{drop}(t) = I(t).R_{pdn} + L_{pdn}.\frac{di(t)}{dt}$$
$$V_{load}(t) = V_{DD} - V_{drop}(t)$$

Waveform Example



Key Point

Dynamic IR Drop causes momentary voltage droop during switching which can cause timing and functional failures.

9. Static IR Drop vs Dynamic IR Drop

Parameter	Static IR Drop	Dynamic IR Drop
Nature	Steady state (DC)	Transient (AC)
Cause	Resistance (R)	Resistance (R) + Inductance (L)
Current	Constant DC current	Time-varying current
Depends on	I and R_{pdn}	I(t), R_{pdn}, L_{pdn}, di/dt
Occurs when	Even without switching	During switching activity
Effect	Reduces effective VDD	Voltage droop / noise
Fix Focus	Reduce resistance	Reduce L and di/dt, Add decaps
Analysis Tool	DC IR Drop analysis	Transient IR Drop analysis
Sign-off Metric	$V_{min} \geq V_{DDmin} - \Delta IR_{static}$	$V_{droop} \leq \Delta IR_{dynamic}$ limit

10. Impact of IR Drop

- Timing degradation → slower circuits.
- Functional failures (wrong logic state).
- Increased leakage and power.
- Possible latch-up and reliability issues.
- Yield loss in manufacturing.

IR Drop Limit

- V_{min} at any point should be within allowed limit.
- Example: $V_{min} \geq 0.9 \times VDD_{nom}$ (Depends on tech).

11. IR Drop Analysis Types

- Static (DC) IR Drop Analysis → consider leakage + dc current.
- Dynamic (AC/Transient) IR Drop Analysis → consider switching.
- Full chip EMIR analysis → IR + EM + Current density.
- Use tools like Voltus, RedHawk, StarRC, Voltus-Fi etc.

12. Good Power Grid Practices

✓ Build strong power grid with rings + straps + mesh.
✓ Use multiple vias (via arrays) at critical points.
✓ Place decaps close to high switching blocks.
✓ Follow DRC, EM and foundry rules strictly.
✓ Always perform IR Drop and EM checks iteratively.
✓ Verify early and fix before sign-off.

Good Power Grid
=
Low IR Drop
+
Robust Silicon
+
Better Yield

13. Practical Observation

- Poor power grid → more IR Drop.
- More IR Drop → more timing failures.
- Proper planning early saves area, power and ECO cost.
- Always verify after any ECO or placement change.

14. Common Mistakes

× Too few straps and vias.
× Long via chains and stubs.
× Ignoring dynamic IR Drop.
× Decaps far from switching blocks.
× Not running EMIR analysis.

15. Interview / Viva Points

Q1. What is IR Drop ?
→ Voltage drop in PDN due to current and resistance / inductance.

Q2. Difference between static and dynamic IR Drop ?
→ Static is DC, dynamic is due to switching and di/dt.

16. Key Takeaways

✓ Static IR Drop → due to R.
✓ Dynamic IR Drop → due to R + L.
✓ Strong PDN reduces both.
✓ Use rings, straps, mesh, vias, decaps.
✓ IR Drop analysis is must before sign-off.

Author: Konda Tharun Vivekananda

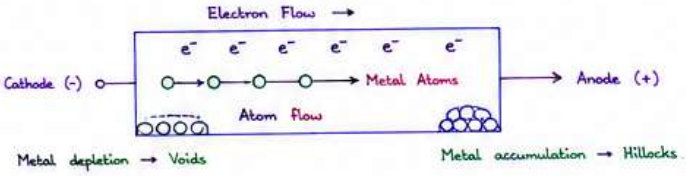
PD Power Planning

Electromigration (EM) Analysis

1. Electromigration – Definition

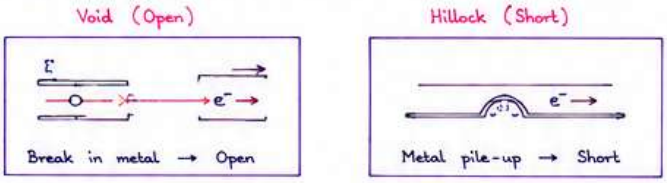
- Electromigration is the gradual movement of metal atoms in a conductor due to momentum transfer from high current electrons.
- It causes voids (open) or hillocks (short) → reliability failure.
- EM mainly affects power grid (rings, straps, mesh, vias).
- EM analysis ensures long term reliability of PDN.

2. Electromigration Mechanism



Atoms move in direction of electron flow (from - to +).

3. EM Failure Examples



4. EM Lifetime Equation (Black's Equation)

$$MTTF \propto \frac{1}{J^n * \exp\left(-\frac{E_a}{kT}\right)}$$

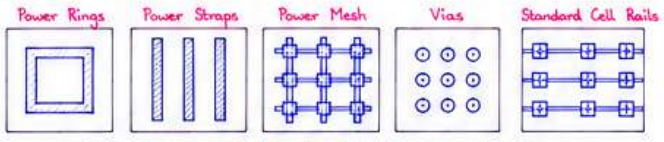
J = Current density (A/cm²)
n = Current density exponent
Ea = Activation energy
k = Boltzmann constant
T = Absolute temperature (K)

⇒ High J, high T → Lower MTTF (Less EM life)
⇒ Lower J, lower T → Higher MTTF (Better life)

5. Factors Increasing EM Stress

① High current density (J).
② Narrow metal width / thin thickness.
③ High temperature.
④ Long unidirectional current path.
⑤ Poor current distribution / imbalance.
⑥ Current crowding at corners, vias, and jogs.
⑦ Small via size or insufficient via arrays.
⑧ Advanced nodes (smaller dimensions → less cross section).

6. EM Affected Structures



All high current PDN elements are prone to EM.

7. Typical Current Density Limits (Reference)

Metal Layer	Typical EM Limit (Jmax) (A/cm²)	Notes
M1 – M2	0.5 – 1.0	Lower layers have tight limits
M3 – M4	1.0 – 1.5	Depends on tech node
M5 – M6	1.5 – 2.5	Wider & thicker metals
M7 and above	2.0 – 4.0	Top metals – Higher limits
Via	0.5 – 1.0	Via EM limit usually lower

Note: Exact values → technology dependent (see foundry PDK).

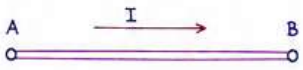
8. EM Fixes / Mitigation Techniques

- ✓ Increase metal width (W) or thickness (T).
- ✓ Add more parallel paths (straps, rings, mesh).
- Add multiple via arrays to reduce via current density.
- Use bigger via size or stacked vias.
- Avoid long continuous unidirectional paths.
- Use redundant power paths for current sharing.
- Avoid sharp corners, use 45° or rounded corners.
- Reduce current crowding using wider connections.
- Balance currents across multiple domains.
- Follow EM rules from foundry strictly.

Key Point
Reduce current density (J) and provide redundant paths to improve EM reliability.

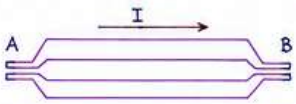
9. Good vs Bad Routing for EM

Bad Routing (High EM Stress)



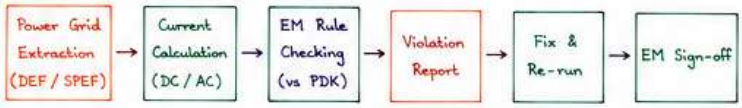
Long, thin and single path
→ High current density
→ Low EM lifetime

Good Routing (Low EM Stress)



Multiple parallel wide paths
→ Low current density
→ High EM lifetime

10. EM Analysis Flow



11. EM Analysis Checks

- Check all metals (rings, straps, mesh, rails).
- Check all vias and via stacks.
- Identify EM hotspots (high J regions).
- Verify against EM limits from PDK.
- Report worst case MTTF and violations.

EM Sign-off Criteria
① No EM violations.
② All J ≤ Foundry limit.
③ MTTF ≥ Required life (e.g., 10 years).
④ Reliable for worst case conditions.

12. EM vs IR Drop – Quick Comparison

Feature	IR Drop	Electromigration
Cause	Resistance (R)	Electron momentum
Effect	Voltage drop	Metal failure (void/hillock)
Depends on	R, length, current	Current density, Temp, Time
Where	All nodes	Power grid (high current)
Analysis	Voltage based	Current density based
Sign-off	Vmin ≥ Vlimit	MTTF ≥ Required life

13. Practical Observation

- EM failures happen slowly → not visible in short term.
- IR Drop causes immediate timing failures.
- Both must be analyzed together for reliable PDN.
- Early planning with proper rings + straps reduces EM risk.

Good Practice
- Design for low J.
- Provide redundancy.
- Follow EM rules.
- Validate early & re-check after ECO.

14. Interview / Viva Points

Q1. What is Electromigration ?
→ Movement of metal atoms due to electron flow causing voids and hillocks.

Q2. What increases EM stress ?
→ High current density, high temp, Long path, small width, etc.

15. Common Mistakes

- × Using single narrow straps.
- × Ignoring via EM limits.
- × Long unidirectional routes.
- × Missing current crowding points.
- × Not running EM after ECO.

16. Important Metrics

- Current Density (A/cm²)
- MTTF (Mean Time To Failure)
- EM Rule Margin (%)
- Worst Case J
- Temperature (°C)

17. Key Takeaways

- ✓ EM is a reliability issue in PDN.
- ✓ High current density reduces metal life.
- ✓ Use wider metals, more paths & vias.
- ✓ Follow foundry EM rules.
- ✓ Always sign-off for EM along with IR Drop.

PD Power Planning

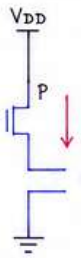
1. Power Consumption Fundamentals – Introduction

- Power is the rate at which energy is consumed in a circuit.
- Power increases with switching activity, frequency, voltage, capacitance and leakage.
- In ICs, power is one of the most critical PPA metrics.
- Total Power = Dynamic Power + Leakage Power + Short Circuit Power

2. Types of Power in VLSI

Power Type	Symbol	Description	Depends on
Dynamic (Switching)	P_{dyn}	Power due to charging/discharging of capacitance during switching.	C_L, V_{DD}, f, α
Short Circuit	P_{sc}	Power due to short circuit current during input transitions.	V_{DD}, t_r, t_f, freq.
Leakage (Static)	P_{leak}	Power due to leakage currents even in steady state.	V_{DD}, Temp, Process
Total Power	P_{total}	Sum of all components.	All factors

3. Dynamic (Switching) Power



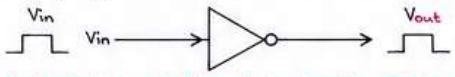
$$P_{dyn} = \alpha C_L V_{DD}^2 f$$

Where,
- α = Activity factor (0 to 1)
- C_L = Effective switching capacitance
- V_{DD} = Supply voltage
- f = Switching frequency

Observations:
- Quadratic relation with V_{DD} → most dominant factor.
- Linear relation with frequency (f) and capacitance (C_L).
- Activity factor depends on design and input vectors.

4. Short Circuit Power

- During input transitions, both NMOS and PMOS conduct simultaneously → creates short circuit path from VDD to GND.
- Short circuit current depends on input rise/fall time (t_r, t_f).
- More in fast switching (small t_r, t_f).
- Generally smaller than dynamic power but increases with high frequency and low V_{DD}.



Short circuit current flows during transition window.

5. Leakage (Static) Power

- Exists even when circuit is not switching.
- Main sources: Subthreshold leakage, Gate tunneling leakage, Junction leakage, BTBT leakage.
- Increases exponentially with temperature.
- Becomes significant in deep sub-micron technologies.

$$P_{leak} = V_{DD} \times I_{leak}$$

Where,
I_{leak} = Total leakage current in the circuit.

6. Total Power Equation

$$P_{total} = P_{dyn} + P_{sc} + P_{leak}$$
$$= \alpha C_L V_{DD}^2 f + P_{sc} + V_{DD} I_{leak}$$

Power $\propto V_{DD}^2 f$ (dominant term → dynamic power)

Key Point
Lowering VDD is the most effective way to reduce power.

7. Factors Increasing Power

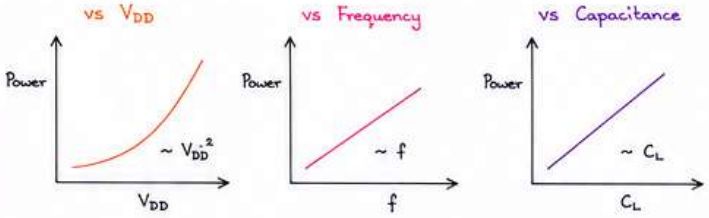
Dynamic Power Factors

1. High switching activity (α ↑)
2. High supply voltage (VDD ↑)
3. High frequency (f ↑)
4. Large load capacitance (C_L ↑)
5. Glitching / unnecessary switching
6. Large fanout
7. Poor clock gating
8. Inefficient logic styles

Leakage Power Factors

1. High temperature
2. Low threshold voltage (V_{th})
3. Process variations
4. Smaller device dimensions
5. High supply voltage
6. Poor power gating
7. Body bias not optimized

8. Power vs Factors (Impact)



9. Power Reduction Techniques

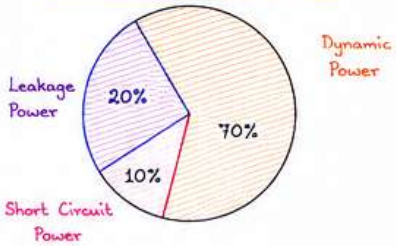
- ✓ Lower supply voltage (DVFS).
- ✓ Reduce switching activity (clock gating, operand isolation).
- ✓ Use multi-threshold cells.
- ✓ Reduce capacitance (optimal sizing, logic optimization).
- ✓ Power gating / sleep transistors.
- ✓ Use efficient clock tree and low glitch design.
- ✓ Use low leakage libraries.
- Body biasing and adaptive threshold.
- Dynamic frequency scaling.

Goal
Minimize Power while maintaining Performance and Reliability.

10. Power Estimation Flow

Netlist + Libs → Vector / SAIF / Activity File → Power Analysis Tool → Power Reports (Breakdown) → Optimization & Re-analysis

11. Power Breakdown Example



Observation
- Dynamic power is usually dominant in most designs.
- Leakage increases in advanced nodes.
- Short circuit is smaller but important in high speed designs.

12. Practical Observation

- Power optimization should start early (architecture + RTL).
- Power Intent (UPF) is essential for low power flows.
- Always validate power with IR Drop, EM and Thermal.
- Power, Performance and Area are always inter-dependent.
- Good power planning (rings, straps, mesh) improves reliability.

13. Interviews / Viva Points

Q1. What are the main components of power?
→ Dynamic, Short Circuit and Leakage.

Q2. Which factor has the most impact on dynamic power?
→ Supply voltage (quadratic relation).

14. Common Mistakes

- ✗ Ignoring leakage power in advanced nodes.
- ✗ Not considering activity factor in estimation.
- ✗ Using too high frequency unnecessarily.
- ✗ Not optimizing VDD and threshold voltages.
- ✗ Ignoring short circuit power in fast designs.

15. Important Metrics

- Total Power (mW)
- Power Density (mW/mm²)
- Leakage Current (nA / µA)
- Activity Factor (α)
- Power Efficiency (P/Perf)

16. Summary / Takeaways

- ✓ Power is critical for performance, reliability and thermal.
- ✓ Dynamic Power dominates in most digital designs.
- ✓ Leakage is critical in advanced technologies.

PD Power Planning

Low Power Techniques – Overview

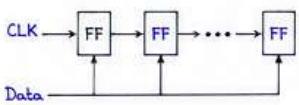
- Low power design reduces power consumption while meeting performance, area and functionality.
- Achieved by reducing switching, leakage, voltage, frequency and power in unused blocks.
- Key techniques in PD Power Planning:

1) Clock Gating 2) Power Gating 3) Multi VDD
4) DVFS 5) Always On Logic 6) Multi Power Domain Design

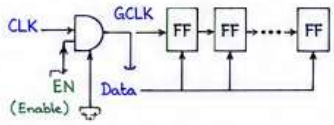
1. Clock Gating

- Disables clock to idle sequential blocks / registers.
- Prevents unnecessary switching of flip-flops.
- Reduces dynamic power significantty.

Without Clock Gating



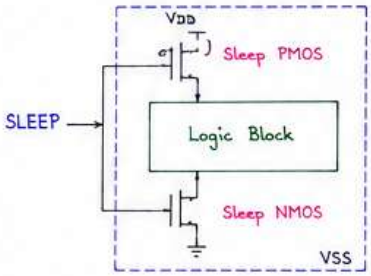
With Clock Gating



Key Points
- ✓ Enable = 0 → Clock OFF → No switching.
- ✓ Enable = 1 → Clock ON → Normal operation.

2. Power Gating

- Turns OFF power supply to idle blocks using sleep transistors.
- Reduces leakage power by cutting VDD to the block.
- Wake-up when block is needed.

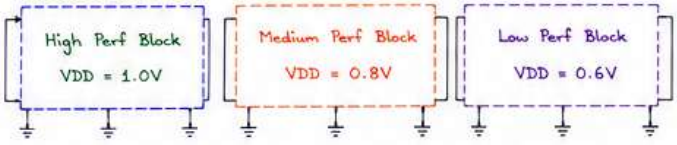


SLEEP	State	Block Power
0	Active	ON
1	Sleep	OFF

Key Points
- ✓ High sleep → Power OFF (VDD isolated).
- ✓ Low sleep → Power ON.

3. Multi VDD (Multi Voltage)

- Different blocks operate at different supply voltages.
- Critical / High perf blocks → High VDD.
- Non-critical blocks → Low VDD.



Advantages
- ✓ Reduces dynamic power ($P_{dyn} \propto V^2$).
- ✓ Improves power efficiency.

Challenges
- × Level shifters required.
- × Different timing constraints.

4. Dynamic Voltage and Frequency Scaling (DVFS)

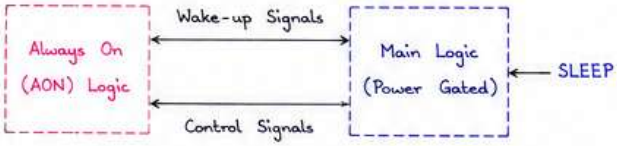
- Adjusts both voltage (VDD) and frequency (f) based on workload.
- Low workload → Lower VDD and f.
- High workload → Higher VDD and f.
- Saves dynamic power and leakage.

Mode	VDD	Frequency (f)	Power
High Perf	1.0 V	1.0 GHz	High
Medium Perf	0.8 V	600 MHz	Medium
Low Power	0.6 V	300 MHz	Low

Key Point
Power $\propto V^2 \times f$
DVFS → adapts power to workload.

5. Always On Logic

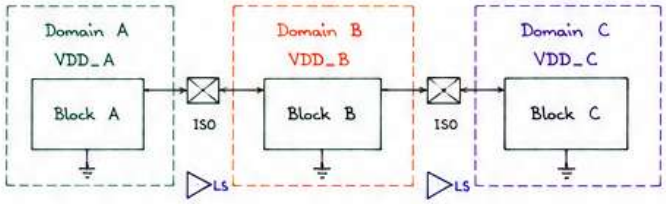
- Small set of logic remains powered even in sleep modes.
- Handles wake-up events, power control, state retention.
- Ensures quick and reliable wake-up.



Examples ✓ Wake-up timer ✓ Retention flops ✓ Power controller

6. Multi Power Domain Design

- Chip is divided into multiple power domains.
- Each domain can be ON/OFF and operate at its own voltage.
- Isolations, level shifters and retention cells ensure safe operation.



ISO : Isolation Cell LS : Level Shifter

7. Comparison of Techniques

Technique	What it Reduces	Power Saved	Overhead	Best Use
Clock Gating	Dynamic Power	High	Low	Idle sequential logic
Power Gating	Leakage Power	Very High	Medium	Large idle blocks
Multi VDD	Dynamic Power	Medium-High	High	Perf criticality based
DVFS	Dynamic Power	High	Medium	Variable workload
Always On Logic	N/A (Support)	N/A	Low	Wake-up & control
Multi Power Domain	Both Dyn & Leak	Very High	High	Complex SoC

8. Practical Observation
- Clock gating is the easiest and most common technique.
- Power gating gives highest leakage reduction but adds wake-up delay.
- Multi VDD and DVFS need careful timing and verification.

9. Benefits
- ✓ Lower power consumption.
- ✓ Reduced heat dissipation.
- ✓ Longer battery life.
- ✓ Better reliability.
- ✓ Supports high performance within power limits.

10. Challenges
- × Area overhead (ISO, LS, sleep transistors).
- × Verification complexity.
- × Wake-up latency.
- × Power supply noise issues.

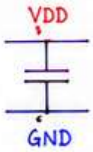
11. Interview / Viva Points
Q1. What is clock gating?
→ It disables clock to idle logic to stop unnecessary switching.
Q2. What is power gating?
→ It turns OFF power supply to idle blocks.

12. Key Takeaways
- ✓ Use clock gating for small idle logic.
- ✓ Use power gating for large idle blocks.
- ✓ Use multi VDD & DVFS for adaptive power.
- ✓ Always On logic is essential for power management.

13. Summary / Revision Points
- Clock Gating → reduce dynamic power by controlling clock.
- Power Gating → reduce leakage by cutting VDD to block.
- Multi VDD → run blocks at different voltages.
- DVFS → scale voltage & frequency based on workload.
- Always On Logic → supports wake-up and control functions.
- Multi Power Domain → independent control of domains for maximum savings.

Decoupling Capacitors (DEcap Cells)

--Definition , function , benifits in power planning.



1. DEFINITION

- Decoupling capacitors (DEcap cells) are MOS capacitors inserted between VDD and GND in the power grid.
- They act as local charge reservoirs to supply / absorb transient current quickly.
- DEcaps are specially designed standard cells with high capacitance density.

2. FUNCTION IN POWER PLANNING

- Provide local energy storage close to standard cells.
- Reduce supply voltage fluctuation during switching.
- Suppress high frequency noise on power rails.
- Improve PDN stability and transient response.
- Reduce IR Drop by supporting instantaneous current.
- Improve timing reliability and reduce glitch / noise induced failures.
- Essential for large designs with high switching activity.

3. TYPES OF DECOUPLING CAPACITORS

① By Implementation :
- MOS DEcaps (On-chip cells)
- MIM Capacitors (Metal-Insulator-Metal)
- MiMCap / MOMCap (Process dependent)

② By Location :
- Core Decaps (Inside core area)
- IO Decaps (Near IO rails)
- Macro Decaps (Around macros / Memories)

4. WHERE TO PLACE DECAPS ?

- Distributed uniformly across the core area.
- Higher density near high switching blocks.
- Near macros, SRAMs, PLLs, SerDes, CPU blocks.
- Avoid placement under large routing congestion.
- Follow foundry recommended guidelines.

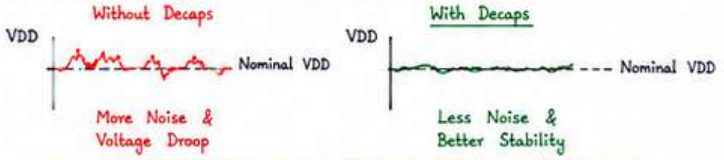
5. DECAP PLACEMENT GUIDELINES

- ✓ Uniformly distribute across core.
- ✓ Maintain good connection to VDD/GND network.
- ✓ Place near power straps / mesh for low impedance.
- ✓ Keep away from blockages & keep routing channels free.
- ✓ Meet target capacitance density (C/Area).
- ✓ Avoid clustering at one location.

KEY POINT

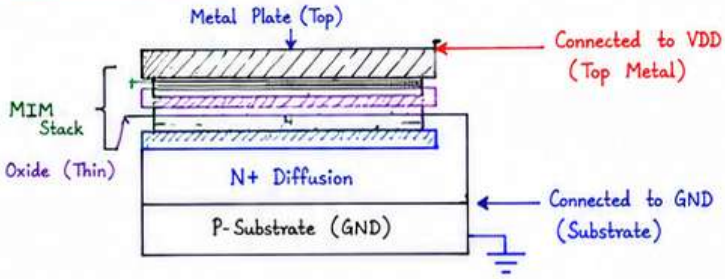
DEcap cells do not replace the power grid. They support the PDN locally by providing instantaneous charge during transients.

6. DECAP EFFECT ON POWER SUPPLY

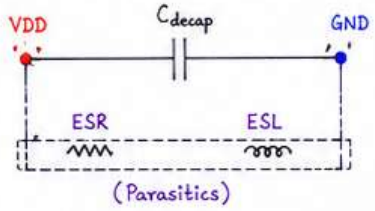


Result : Better voltage stability → Lower IR Drop → Better timing & reliability.

7. DECAP CELL STRUCTURE (MOS DECAP)



8. EQUIVALENT CIRCUIT



Model Parameters
- Capacitance (C)
- ESR (Equivalent Series Resistance)
- ESL (Equivalent Series Inductance)
- Leakage Current

9. BENEFITS OF DECAPS

- ✓ Improve power integrity & voltage stability.
- ✓ Reduce IR Drop and power supply noise.
- ✓ Improve timing closure and reduce setup/hold risk.
- ✓ Reduce dynamic supply bounce.
- ✓ Support high switching activity blocks.
- ✓ Improve chip reliability and reduce soft errors.
- ✓ Help in passing power integrity sign-off (IR/EM).

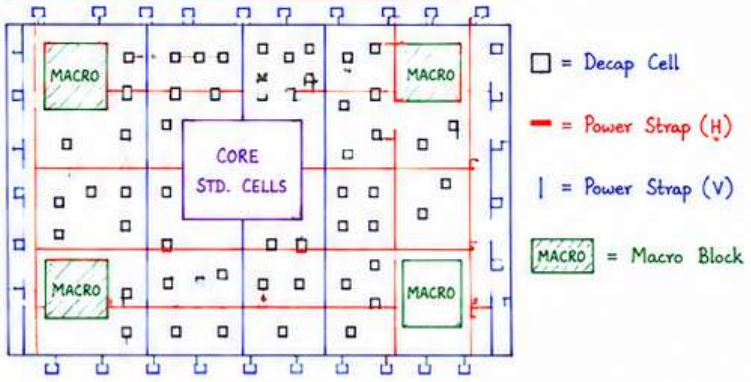
10. DECAP DENSITY (RULE OF THUMB)

$$\text{Capacitance Density (C/Area)} = \frac{\text{Total Decap Capacitance (F)}}{\text{Core Area } (m^2)}$$

Typical target values (Process dependent) :

Technology Node	Cap Density (Approx)
28nm and above	1 ~ 2 nF / mm^2
16nm	2 ~ 4 nF / mm^2
7nm and below	4 ~ 10 nF / mm^2

11. DECAP PLACEMENT EXAMPLE (Top View)



12. PRACTICAL OBSERVATIONS

→ Too few decaps → High IR Drop & voltage droop.

→ Too many decaps → Area overhead & routing blockage.

→ Correct placement & density → Best results.

→ Always follow foundry PDK & power integrity guidelines.

13. VIVA / INTERVIEW POINTS

Q1. Why are decaps required in power planning ?

Ans: To provide instantaneous charge during switching transients, reduce IR Drop, and maintain stable power supply.

Q2. Do decaps replace the power grid ?

Ans: No. Decaps support the PDN locally. The power grid provides the bulk current.

14. SUMMARY POINTS

- Decaps store charge locally between VDD & GND.
- They reduce noise, IR Drop and supply bounce.
- Improve timing, reliability and power integrity.
- Proper placement & density are critical.
- Essential sign-off metric: IR Drop.

Author : Konda Tharun Vivekananda

Power Integrity = Stable PDN + Decaps + Good Grid

PD Power Planning

Decap ...

All Guidelines need to follow in all stages of Power Planning

1. Introduction

- Power planning must follow a set of guidelines at every stage to ensure Low IR Drop, Good EM reliability, Stable Power delivery & Sign-off closure.
- These guidelines apply from Floorplan read to Final Sign-off and must be followed consistently.

2. Guidelines - Overview (Applicable to all stages)

1. Meet Target IR Drop across all Voltage Domains.
2. Satisfy Electromigration (EM) reliability.
3. Maintain Current Density within limits.
4. Ensure Uniform Power Distribution (No hotspots).
5. Minimize / Avoid Power Grid Degradation.
6. Provide Redundancy in critical power paths.
7. Verify Connectivity across rings, straps, vias & rails.
8. Maintain Power Integrity for All modes & corners.
9. Follow Foundry / PDK / Tech File Guidelines.
10. Perform Iterative Analysis & Optimization.

3. Guidelines in each stage

A. Floorplan read & Power estimation

- Understand block power numbers correctly.
- Ensure correct Voltage Domain classification.
- Estimate initial IR Drop & Current requirements.
- Decide Power Domains, Ring locations & Pad/Bump plan.

B. Power Domain Planning

- Keep power domains balanced (Size vs current).
- Avoid too many small domains.
- Place isolation cells & level shifters at domain boundaries.
- Ensure domain switches & always-on strategy.

C. Power Ring Creation

- Use sufficient ring width (Based on current).
- Maintain correct spacing from macros / die boundary.
- Use multiple rings for high current blocks.
- Follow min. metal width & spacing rules.

D. Power Strap Creation

- Distribute straps uniformly across core.
- Follow recommended pitch & width.
- Use multiple layers (H & V) for better distribution.
- Avoid strap blockage of routing channels.

E. Power Mesh Creation

- Ensure mesh density is sufficient for IR Drop target.
- Use appropriate metal layers (Top metals).
- Mesh width & pitch as per current density limits.
- Avoid long narrow segments.

F. Via Reinforcement

- Use multiple vias in parallel for high current.
- Follow recommended via rules (stacked vias).
- Ensure via redundancy in critical connections.

G. Decoupling Capacitor Placement

- Place decaps near high switching activity blocks.
- Use mix of values (Density as per tech guideline).
- Connect decaps to power grid with multiple vias.

Summary Points

- Power planning guidelines must be followed in all stages.
- Main goals are Low IR Drop, EM reliability & Power Integrity.
- Apply guidelines during Rings, Straps, Mesh, Vias & Decaps.
- Always run IR Drop & EM analysis after each iteration.
- Final sign-off ensures stable & manufacturable power grid.

4. Guidelines in each stage (Contd...)

H. Connectivity Verification

- Ensure all rings, straps, mesh & pads are connected.
- Run PG Connectivity checks. (No floating nets)
- Verify domain connections & power switches.

I. IR Drop Analysis

- Verify IR Drop for all domains & corners.
- Meet target IR Drop (e.g., < 5% of VDD / Vtarget)
- Fix hotspots & high IR Drop regions.

J. EM Analysis

- Verify current density on all grid segments.
- Meet EM limits as per tech file.
- Fix EM violations by width / via / layer adjustments.

K. Power Grid Optimization

- Iterate ring / strap / mesh / decap parameters.
- Re-run IR / EM after every change.
- Achieve best trade-off: IR Drop, EM, Area & Timing.

L. Power Plan Sign-off Checks

- IR Drop ✓
- EM ✓
- Connectivity ✓
- No floating power nets.

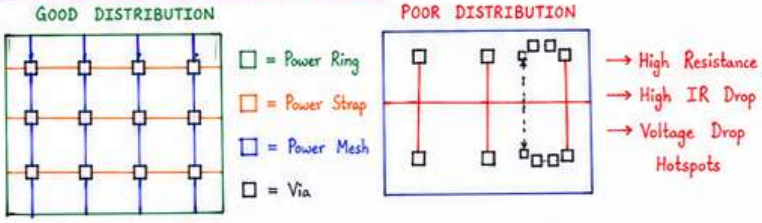
Key Point

Power planning is not a single step activity. It is an iterative process followed across all stages with proper analysis & optimization.

5. Power Grid Quality Checks

Check Item	Target / Limit	Purpose
IR Drop	≤ 5% of VDD (Typical)	Stable Voltage Supply
Current Density	≤ Jmax (Tech File)	EM Reliability
EM Violations	0 (Zero)	Long Term Reliability
Connectivity	100% Connected	No Floating Nets
Decap Density	As per Tech Guide	Reduce Transient Noise
Via Redundancy	As per Rules	Low Resistance Path
Power Bounce	Within Limit	Power Integrity
Hotspot Voltage	Within Spec	Uniform Distribution

6. Uniform Distribution Concept



7. Practical Observations

- ✓ Power planning mistakes are very costly to fix later.
- ✓ Follow guidelines early to avoid iterations.
- ✓ Use adequate margins for EM & IR Drop.
- ✓ Higher level metals are more efficient for distribution.
- ✓ Always validate with real analysis, not assumptions.

8. Viva / Interview Points

Q1. Why guidelines are important in power planning ?

Ans: They ensure reliable, safe & stable power distribution with no sign-off failures.

Q2. What happens if IR Drop is high ?

Ans: It causes functional failures, timing issues & reliability problems.

Don't Forget

- ✗ Don't ignore small IR Drop hotspots.
- ✗ Don't create narrow straps / insufficient vias.
- ✗ Don't leave floating power nets.
- ✗ Don't skip EM analysis.
- ✗ Don't reduce decap density below guideline.

Rule of Thumb

Better Power Grid
→ Lower IR Drop
→ Lower Noise
→ Higher Reliability
→ Successful Sign-off

Author : Konda Tharun Vivekananda

Good Power Planning Today → Successful Silicon Tomorrow

VDD

VSS

INPUT FROM SYNTHESIS
- Netlist
- Libraries
- Constraints

DESIGN UNDERSTANDING
- Block Analysis
- I/O Analysis
- Power Intent

POWER PLANNING
- PDN Design
- IR Drop Control
- EM Reliability

FLOORPLAN CREATION
- Die & Core Area
- I/O Placement
- Power Plan

PLACEMENT
- Macro Placement
- Blockage / Keepout
- Channel Planning

CHECKS & VALIDATION
- DRC / LVS
- IR Drop
- EM / Antenna

READY FOR NEXT STAGE

VDD

PD POWER PLANNING

THANK YOU!

Your time, attention and effort in learning
PD Power Planning concepts are greatly appreciated.

— Keep Exploring. Keep Innovating. Keep Building the Future! —

KEY TAKEAWAYS

- Robust PDN ensures Stable Voltage
- Low IR Drop leads to Better Performance
- EM Reliability ensures Long Term Stability
- Optimized Power Planning improves Timing & Yield
- Early Checks & Validation ensure Signoff Success

A Good Power Plan is the Backbone of a Successful Chip.

Plan Smart. Deliver Power. Ensure Reliability.

Author :
Konda Tharun Vivekananda

